

ECE 105: Introduction to Electrical Engineering

Lecture 1

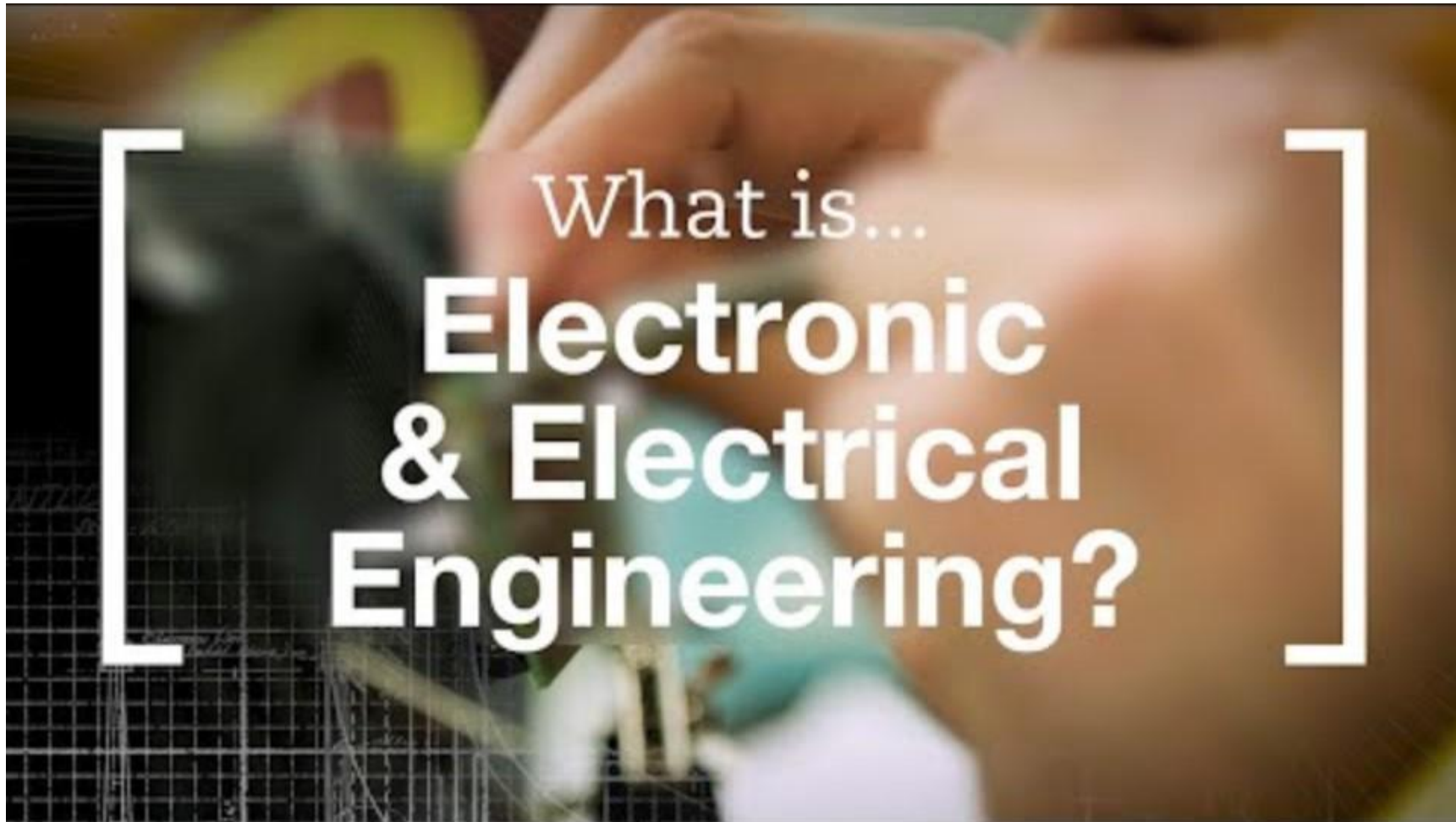
Introduction to EE, and ECE 105

Yasser Khan

Rehan Kapadia

What is Electrical Engineering?

- Electrical Engineering is an engineering discipline concerned with the study, design, and application of equipment, devices, and systems which use electricity, electronics, and electromagnetism.



Let's look at EE at play: 3 little pigs demo by Neuralink



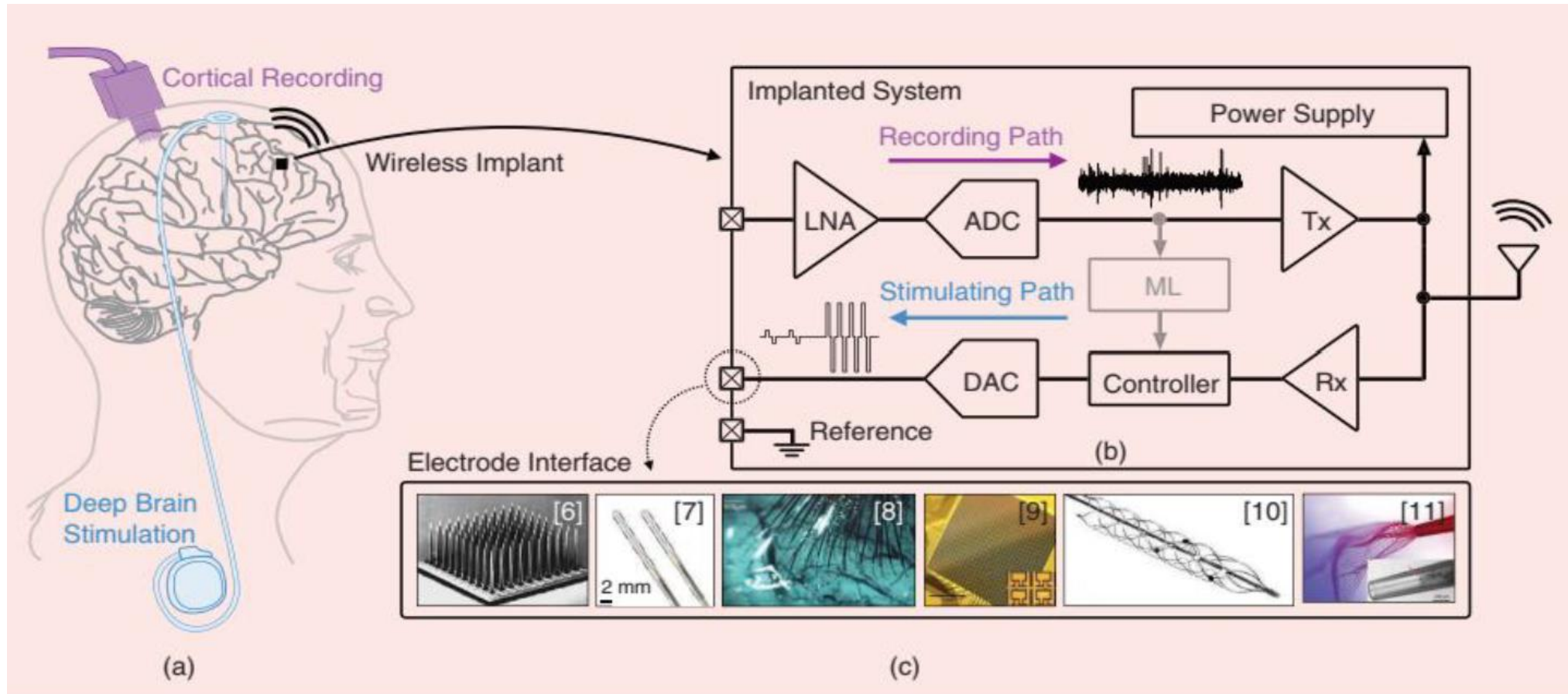
Neuralink – the neural implant



- Every step of the neural implant design and fabrication requires electrical engineering

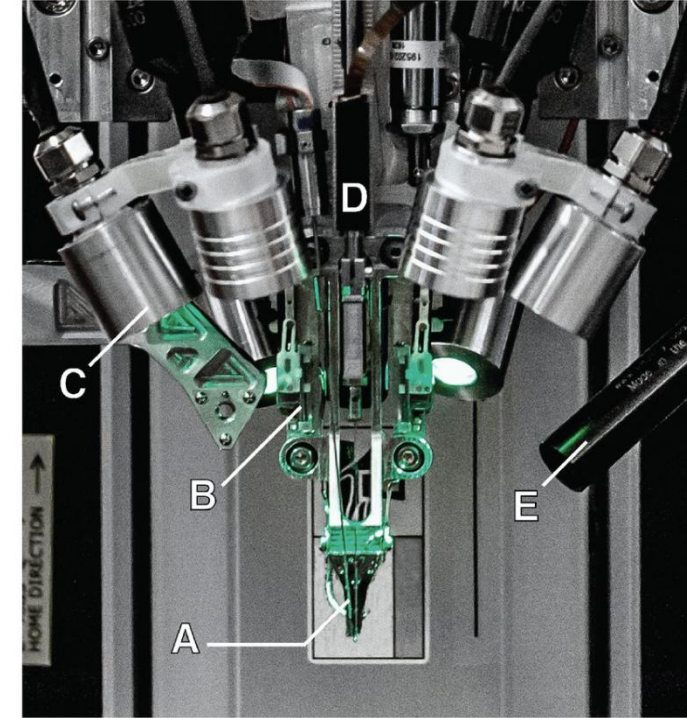
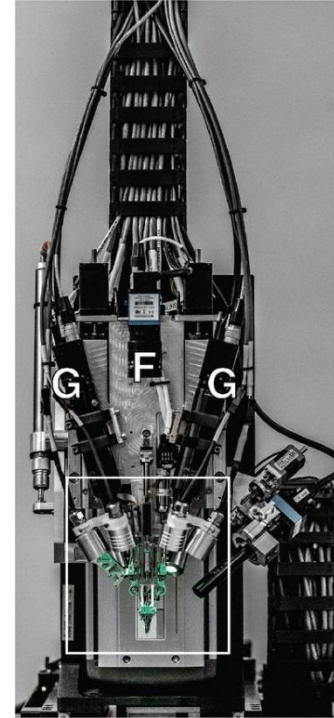
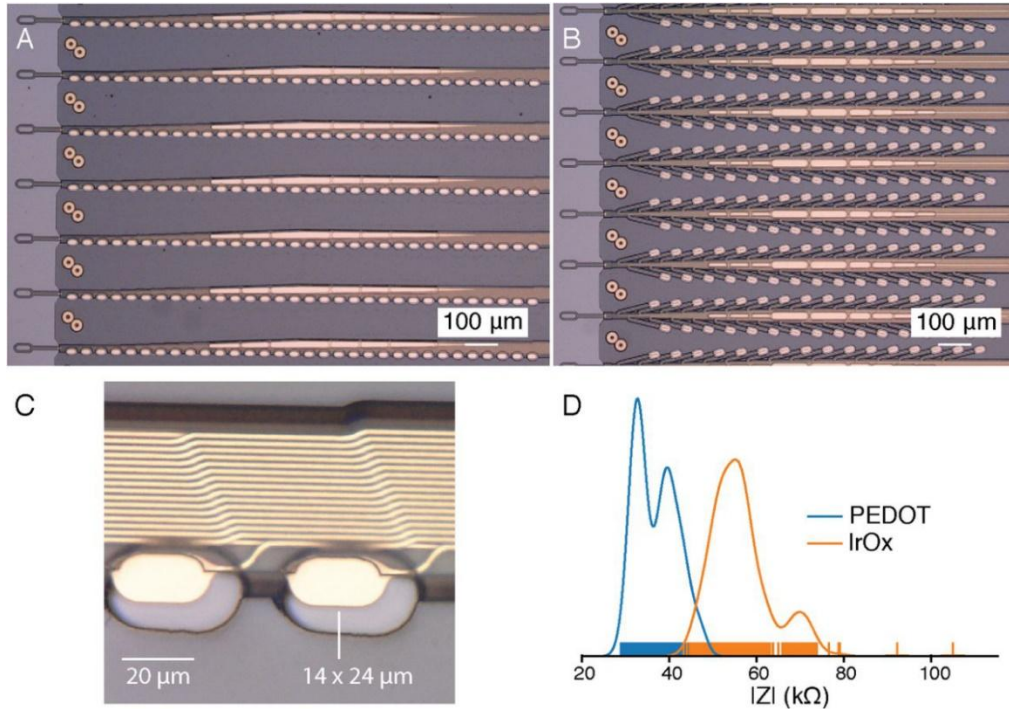


Neural interfaces – under the hood



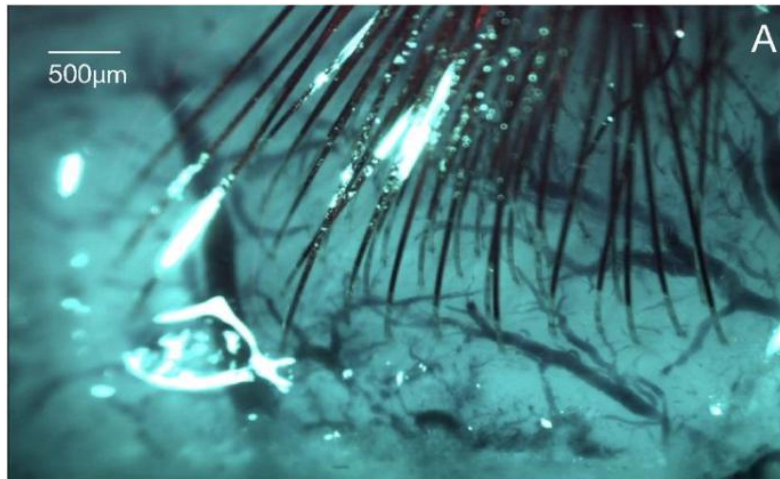
(a) Two clinical neural interfaces: a percutaneous neural recording implant, and a deep brain stimulator. (b) Components for an implantable bidirectional (black) or closed-loop (black and gray) wireless neural interface. (c) Example neural interface electrodes. [From left to right (adapted from Utah microelectrode array [6], SENSIGHT by Medtronic [7], ultrathin polymer threads by Neuralink [8], high-density neural matrix [9], Stentrode by Synchron [10], and ultraflexible syringe-injectable mesh [11]).] LNA: low-noise amplifier; ADC: analog-to-digital converter; Tx: transmitter; ML: machine learning; DAC: digital-to-analog converter; Rx: receiver.

Surgical robots need robotics, **electrical engineering**,
mechanical engineering, control engineering



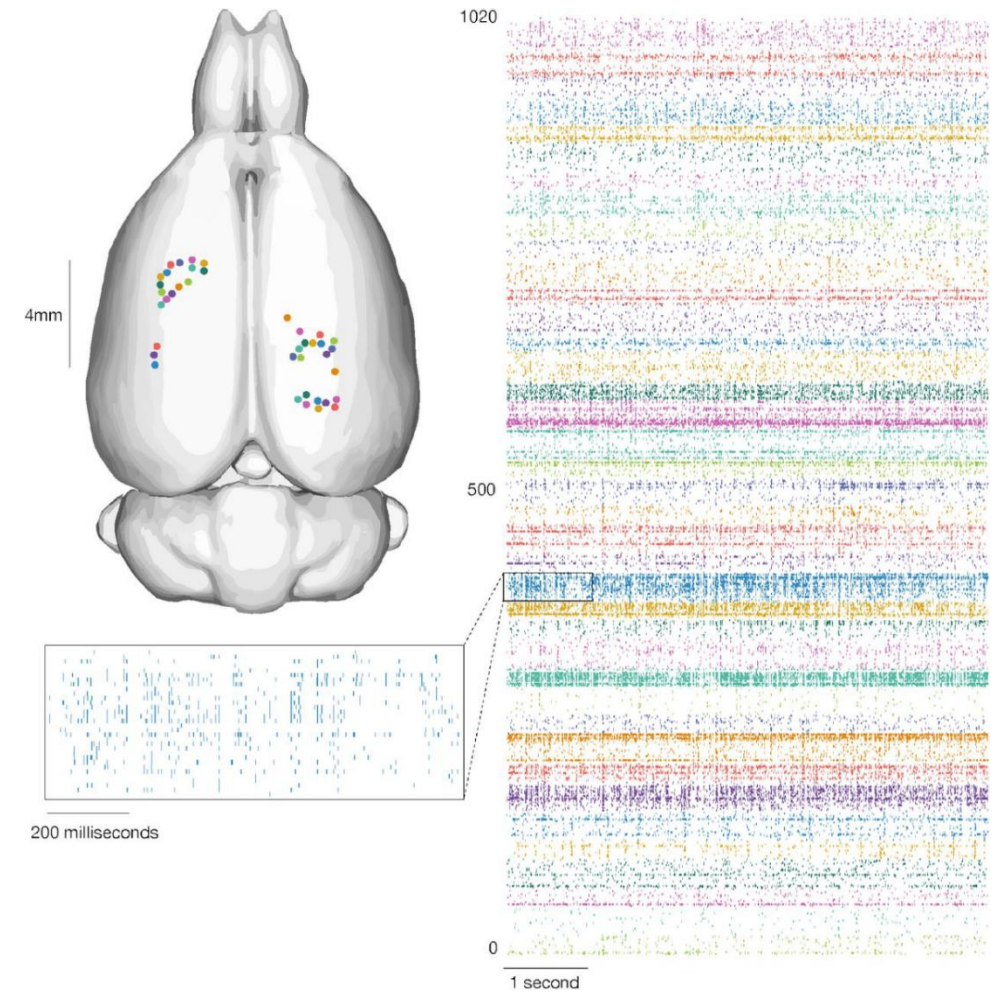
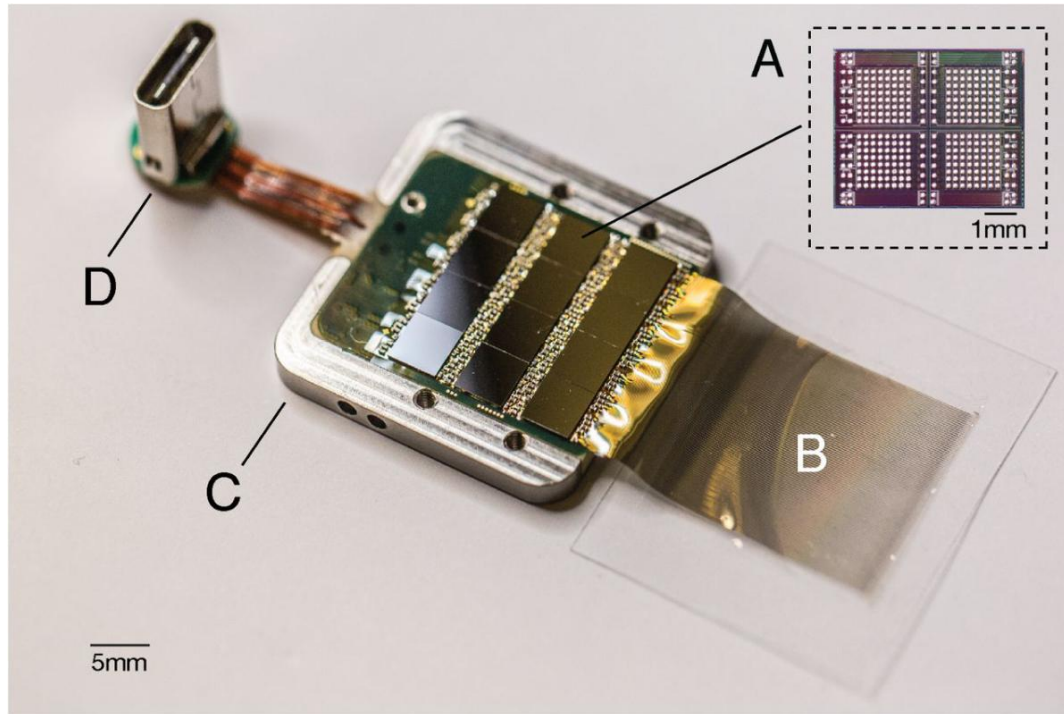
Sensors
need

electrical engineering,
mechanical engineering,
materials science,
chemical engineering



Implants need **electrical engineering**, computer science

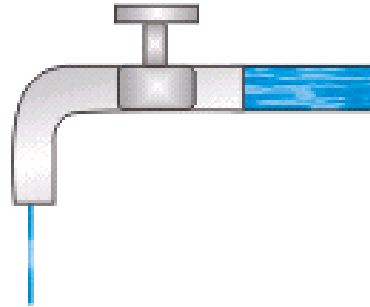
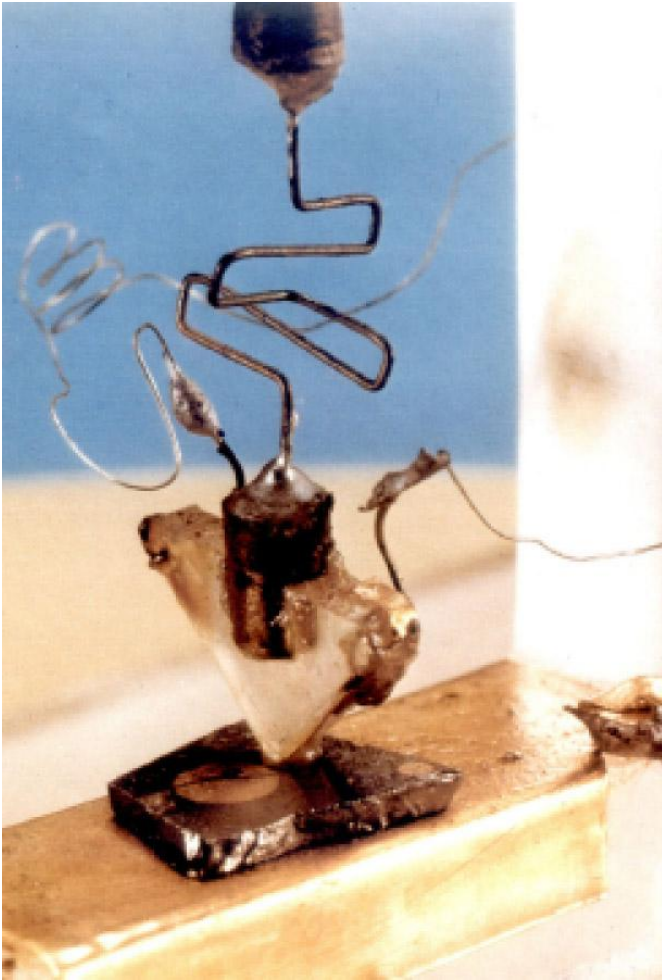
Brain recording from 1024 analog pixels



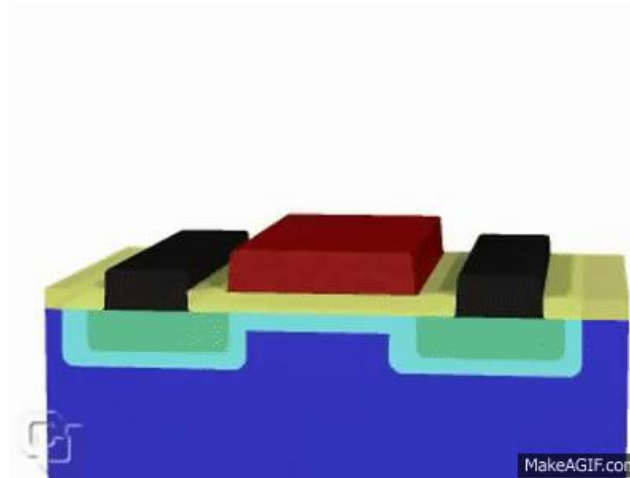
- Each neural recording unit requires sensors, integrated circuits, communications, and power

What is currently driving computing?

How do we carry out so much computing?



We use
transistors!



1956 Physics Nobel Prize

"for their researches on semiconductors and their discovery of the transistor effect"



William Bradford Shockley

🕒 1/3 of the prize

USA

Semiconductor
Laboratory of Beckman
Instruments, Inc.
Mountain View, CA, USA



John Bardeen

🕒 1/3 of the prize

USA

University of Illinois
Urbana, IL, USA



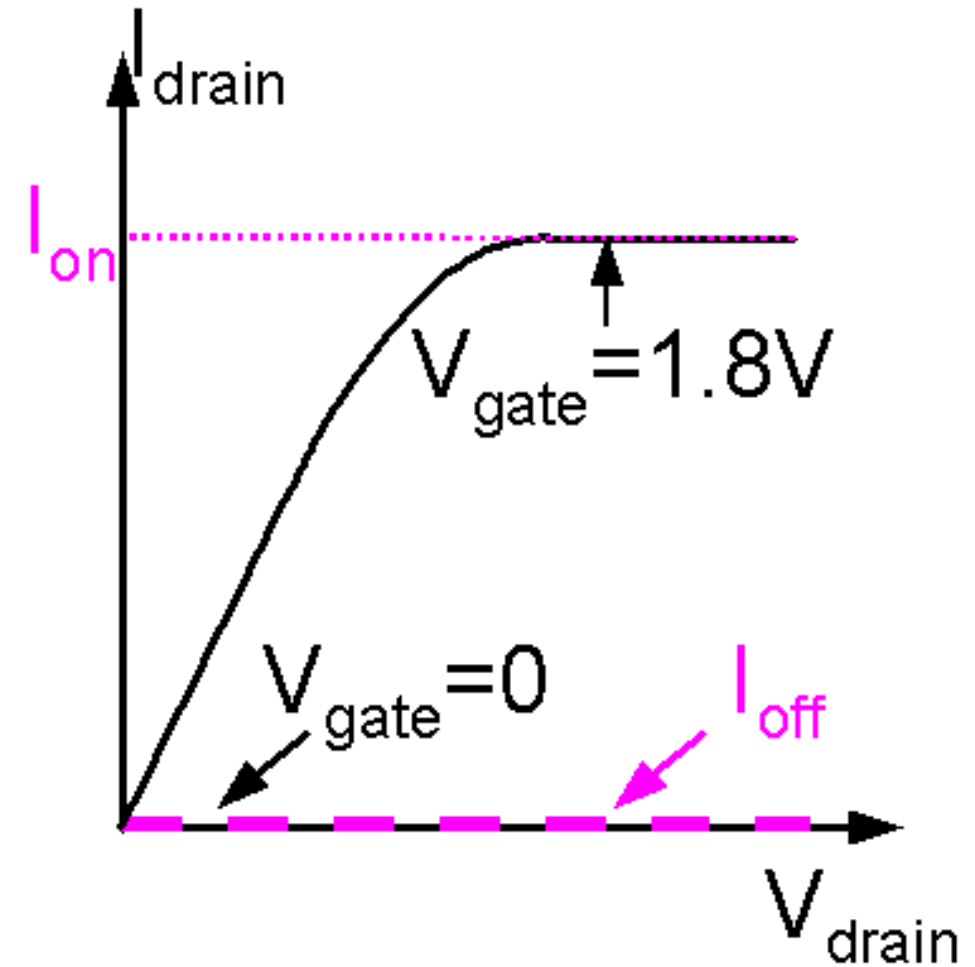
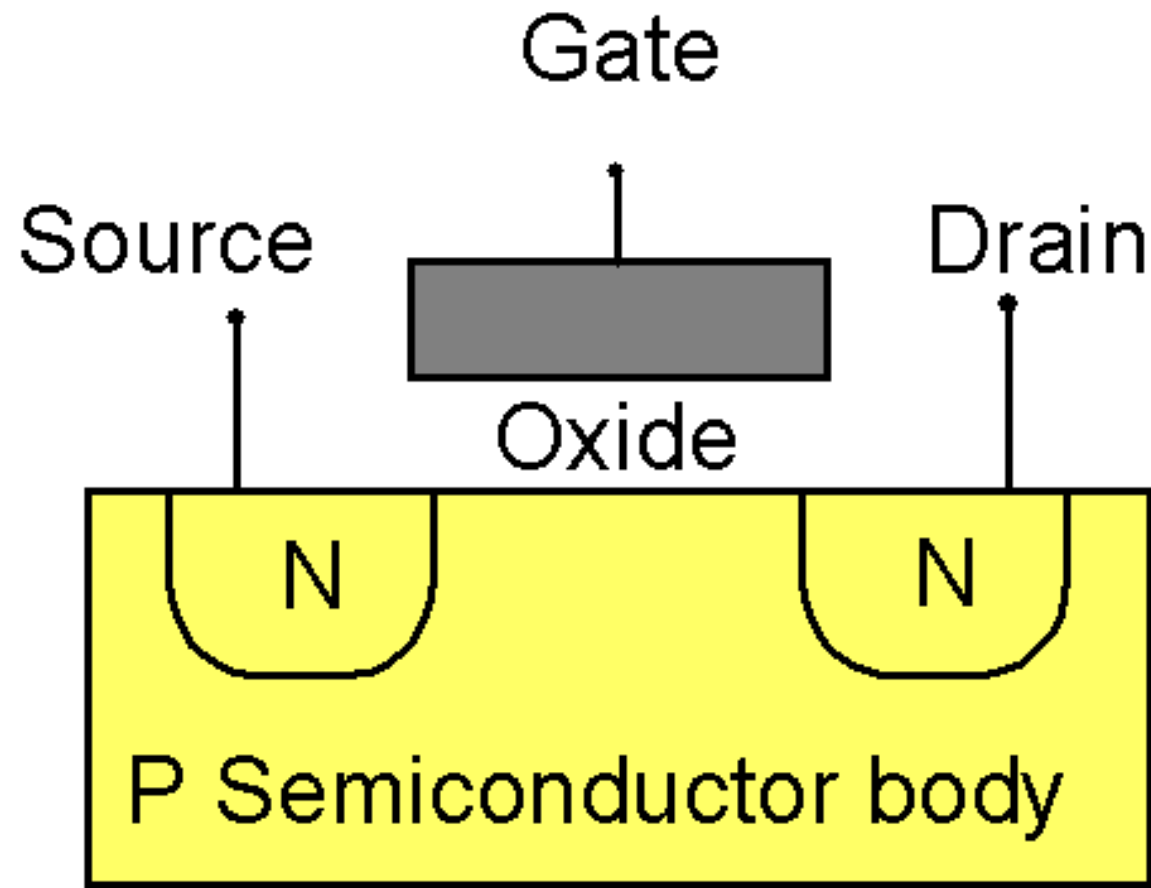
Walter Houser Brattain

🕒 1/3 of the prize

USA

Bell Telephone
Laboratories
Murray Hill, NJ, USA

What is a transistor?



What is desirable: large I_{on} , small I_{off}

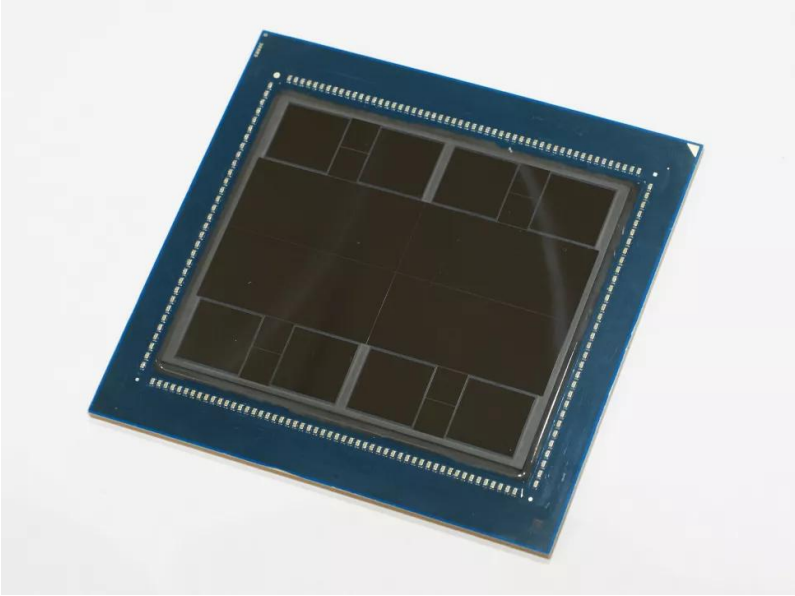
What is a transistor?

1. It acts as an on-off switch.
2. We want to switch it on and off as fast as possible.
3. We want to switch it on and off with as little power usage as possible.

What is desirable: large I_{on} , small I_{off}

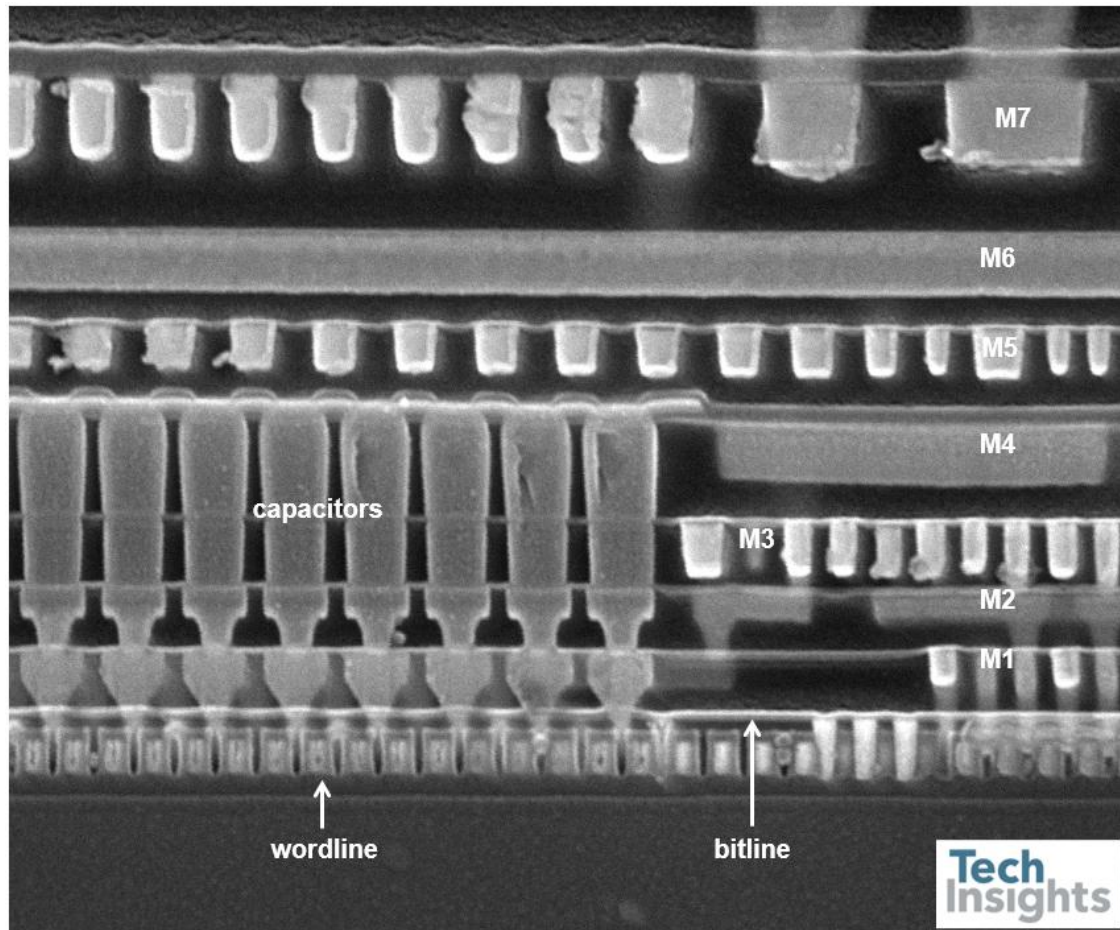
What do modern day integrated circuits look like?

AMD Instinct MI300 Accelerator



- Key points
 - Total of ~150 billion transistors
 - Uses 5 nm node technology
 - Density of transistors is ~130-230 million transistors/mm²
 - In the area of a cell , we could fit ~100k - 1 million transistors
- Multiple parts of this processor use 3-D stacked dies
 - Enables computing, memory, sensing, and stimulation to be integrated together
- Performance ~8x improved over previous MI250 Accelerator
 - If they upgraded the fastest supercomputer to this part, they could train the largest AI/ML systems

What do these look like?

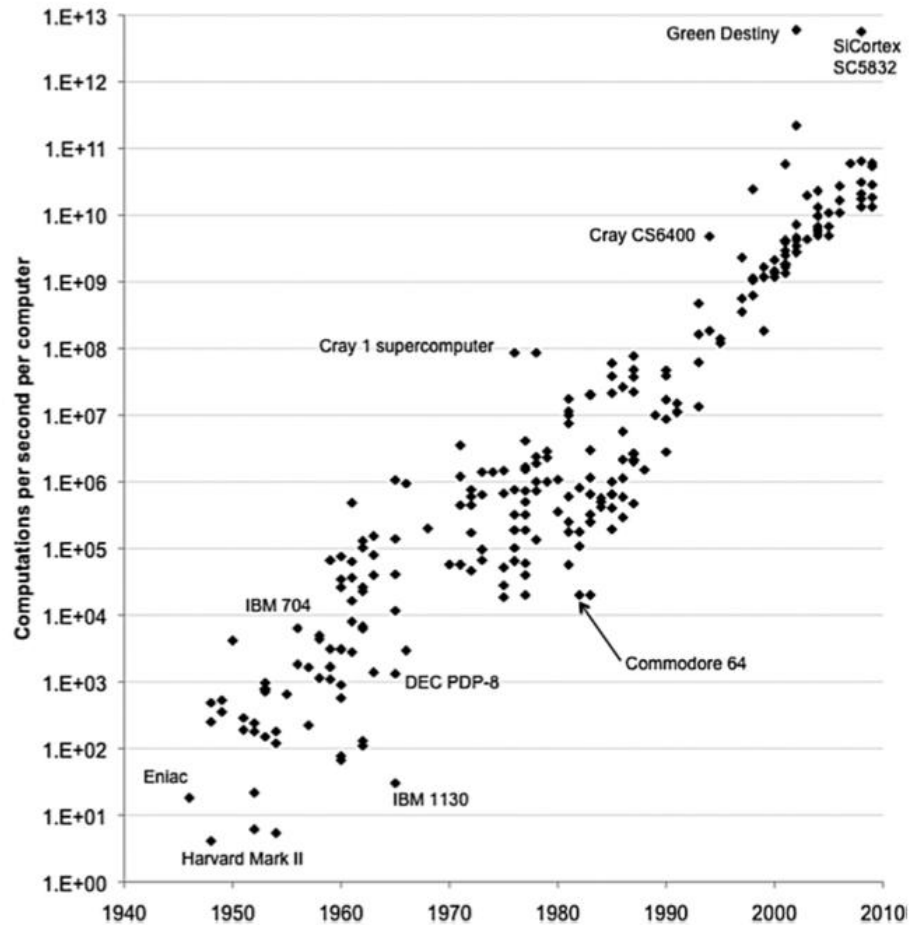


Wires to connect all the transistors together

Active devices (silicon transistors)

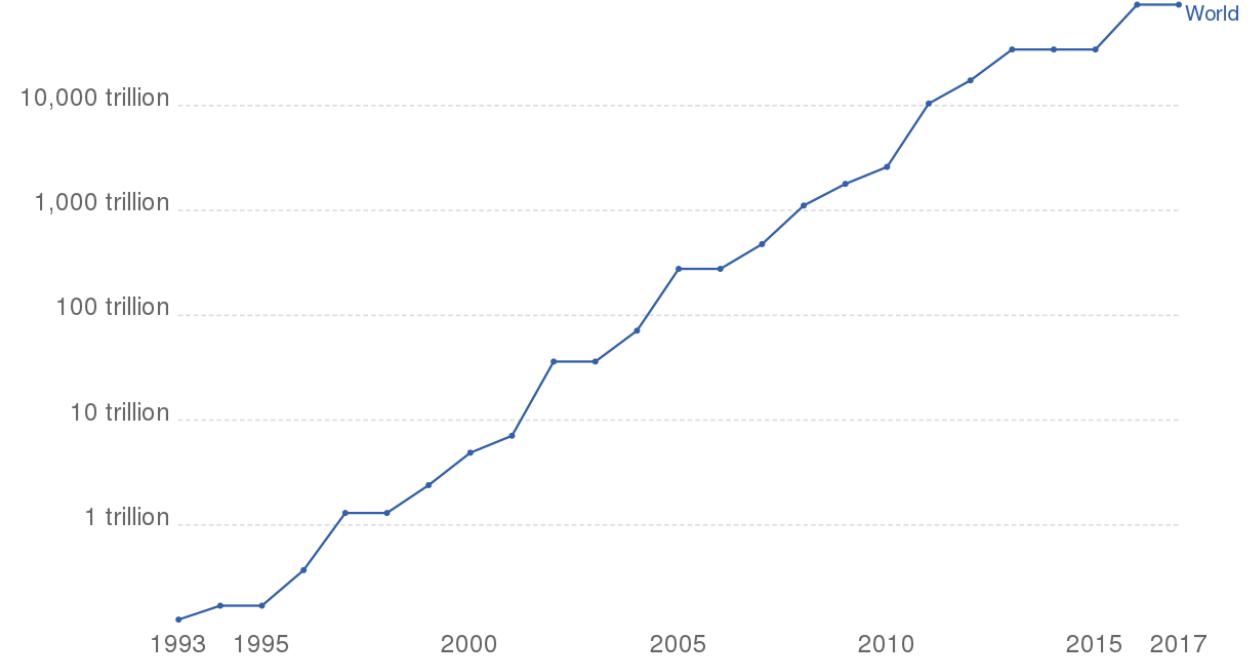
How did we get here?

How much has computing power increased?



Supercomputer Power (FLOPS)

The growth of supercomputer power, measured as the number of floating-point operations carried out per second (FLOPS) by the largest supercomputer in any given year. (FLOPS) is a measure of calculations per second for floating-point operations. Floating-point operations are needed for very large or very small real numbers, or computations that require a large dynamic range. It is therefore a more accurate measure than simply instructions per second.



Source: TOP500 Supercomputer Database

- From 1990 to today, the worlds largest super computers have increased computing power by 100,000x
 - That's a lot!

The First Moon Landing Was Achieved With Less Computing Power Than a Cell Phone or a Calculator



On board Apollo 11 was a computer called the Apollo Guidance Computer. It had 2,048 words of memory that could be used to store “temporary results”—data that is lost when there is no power. This type of memory is referred to as RAM (random access memory). Each word comprised 16 binary digits (bits), with a bit being a zero or a one. This means that the Apollo computer had **32,768 bits of RAM memory**.

My computer has 64,000,000,000 bits of RAM memory

That is 2,000,000 times more than what was needed in Apollo 11.

How do we make transistors faster?

Scaling – How much have we scaled devices?

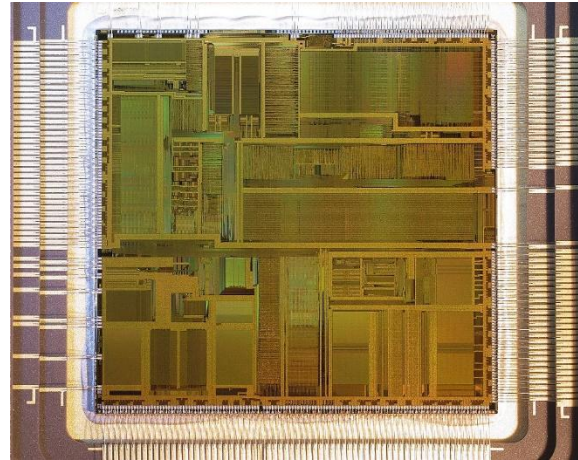
- 1971: The first commercial microprocessor...Intel 4004
 - 4-bits
 - Less than 1 square inch in area
 - Transistors that were 10 micrometers long
 - ~750 kilohertz speed
 - ...but more powerful than the computers from earlier that weighed 30 tons



Let's say that this is a scaled version of how long a transistor was in 1971

Scaling – How much have we scaled devices?

- 1990s: The first Pentium processor
- 66 MHz. That's 100 times faster than in 1970.
 - Can you think of anything else that has improved by 100x in 20 years?

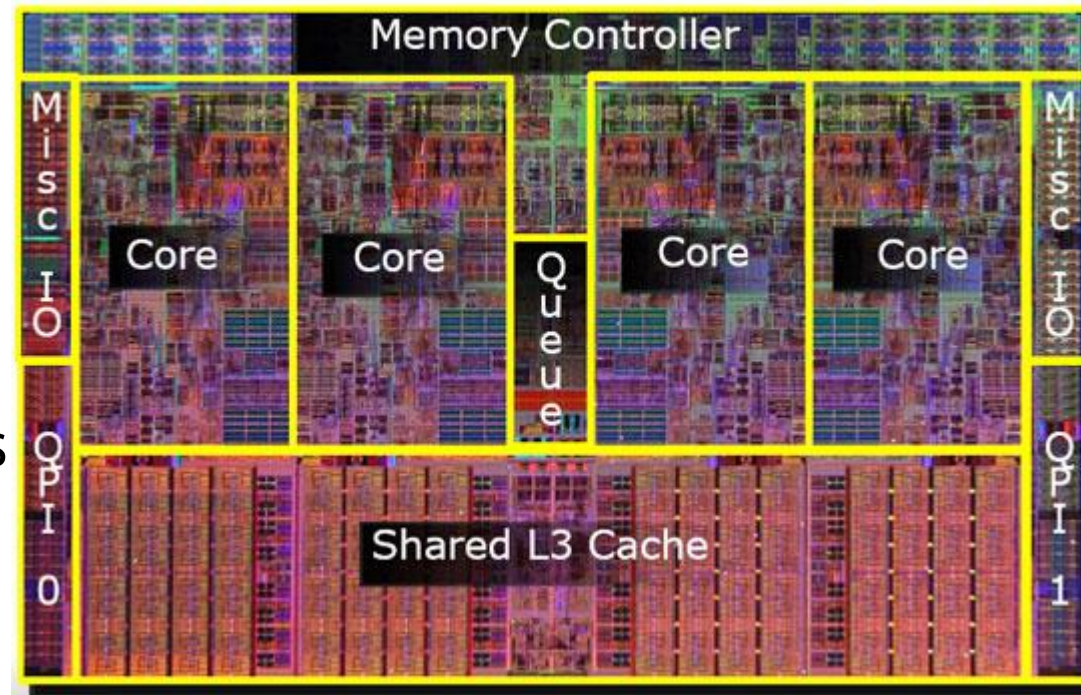


1971

1993

Scaling – How much have we scaled devices?

- 2012:
 - 1-3 GHz
 - Multi-core
 - Super-fast
 - Each transistor is almost 1000 times smaller!

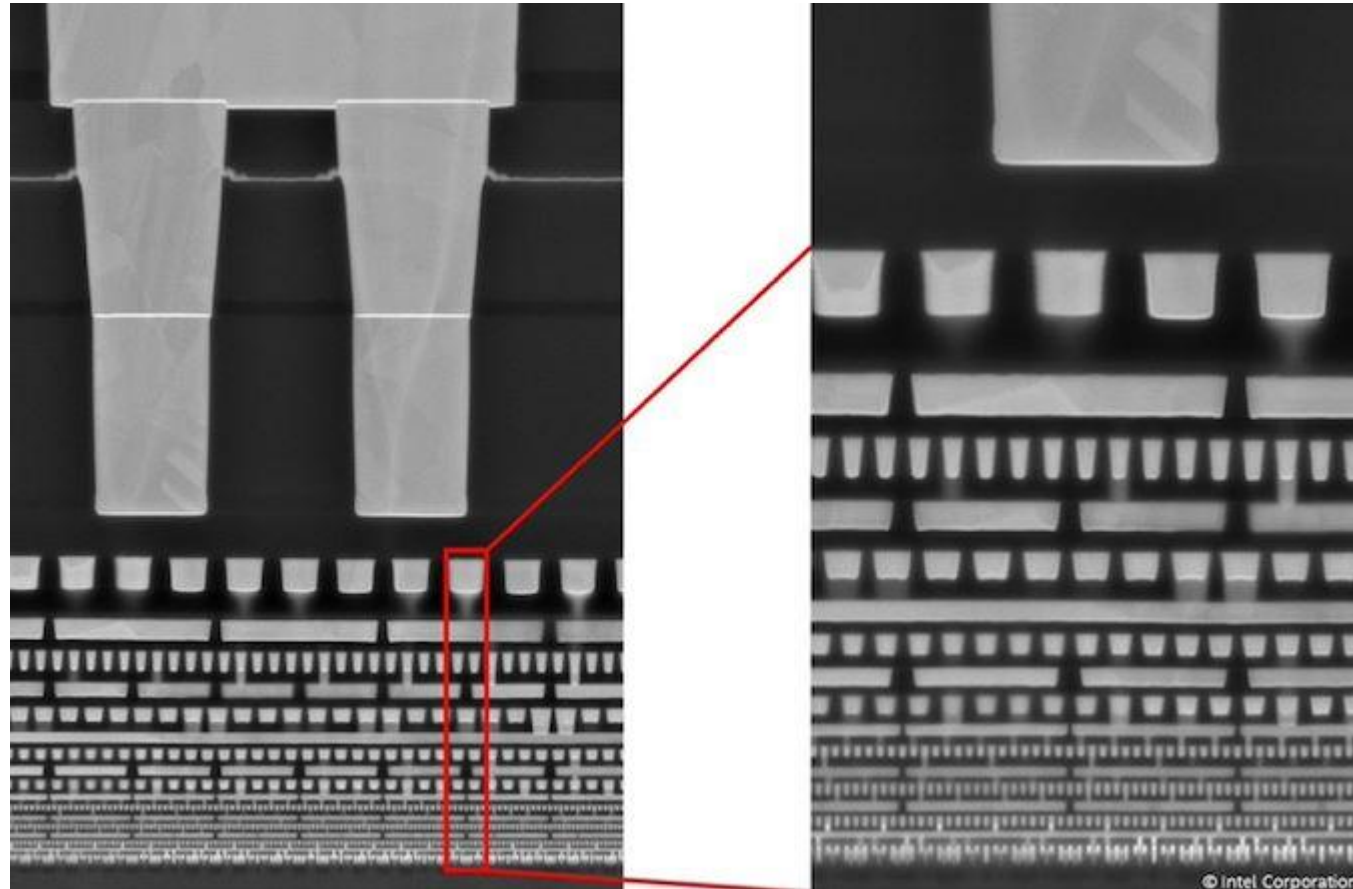


1971

1993

Can't even see the size on the screen...not enough pixels

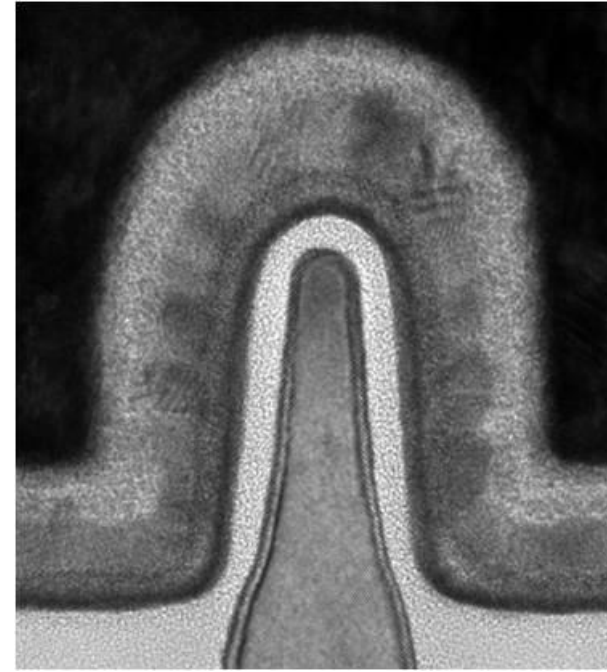
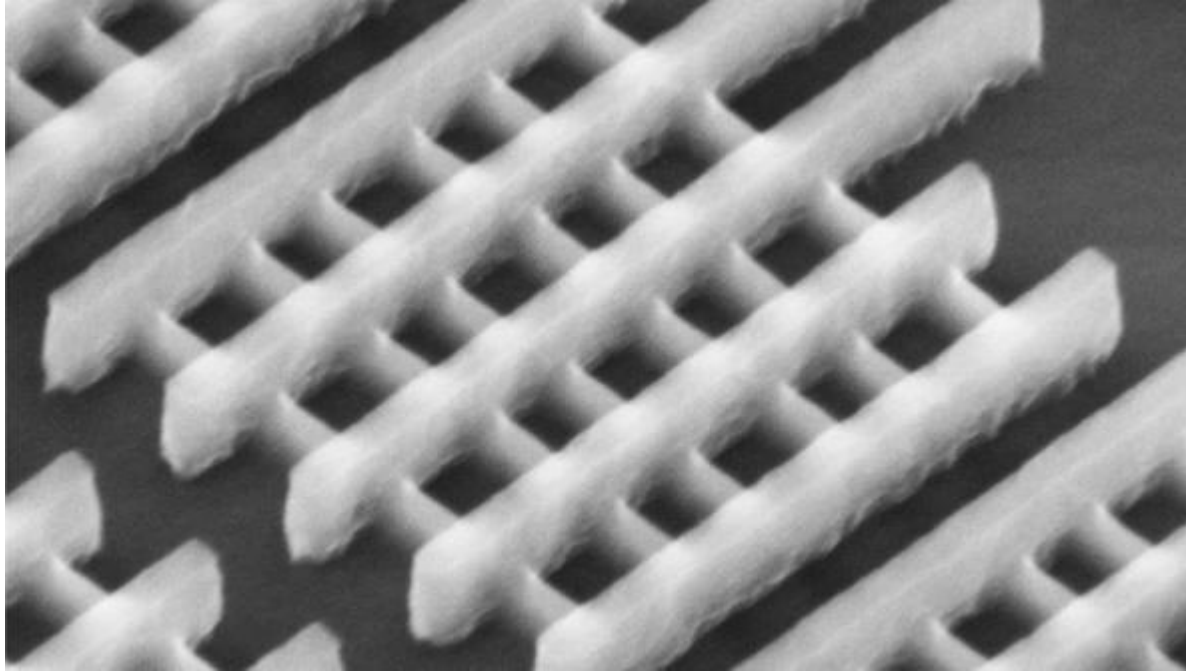
Scaling – How much have we scaled devices?



Intel 4 cross
section: to be
released 2023

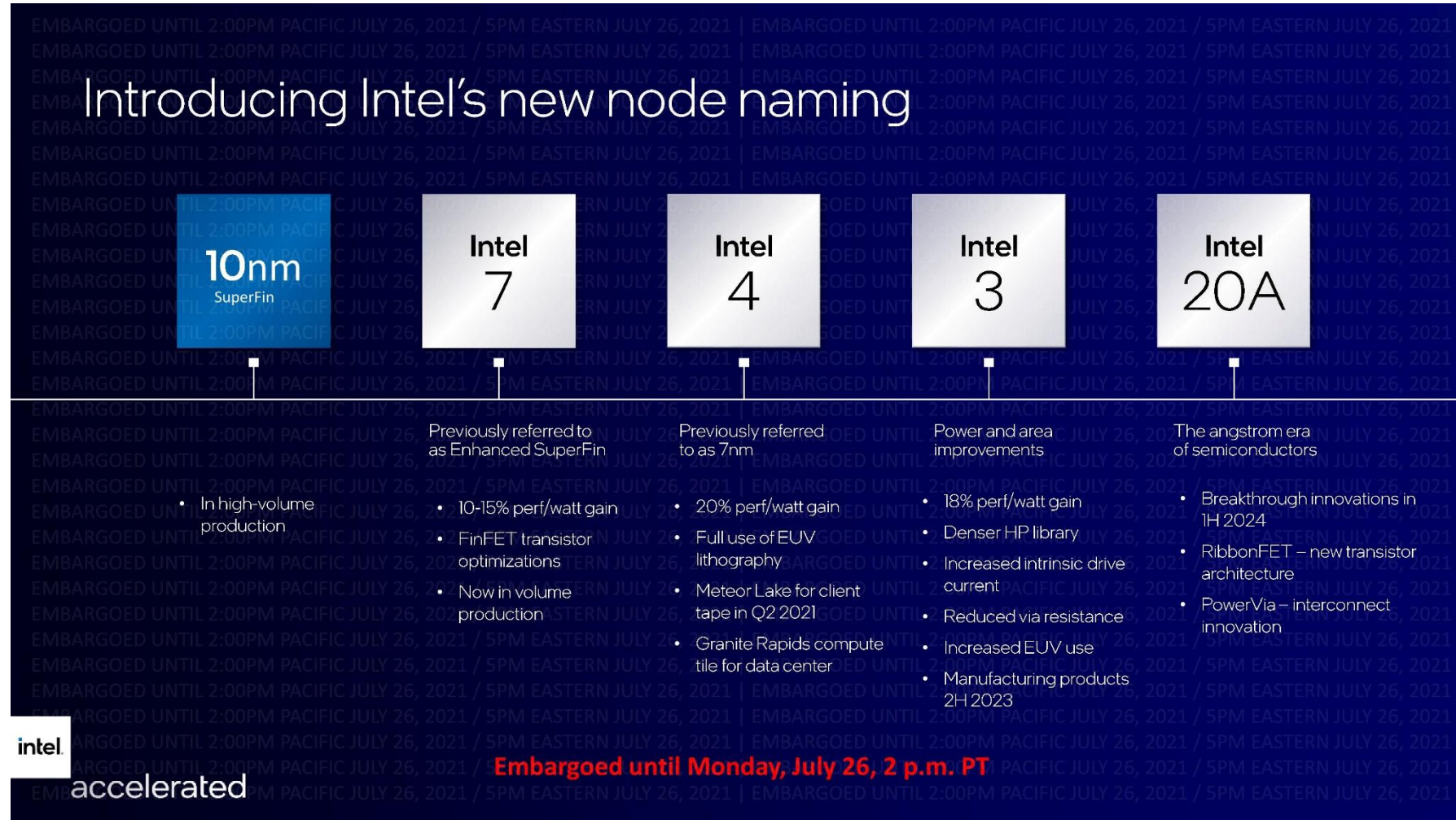
- What does a modern CPU cross section look like?
 - Where are the transistors?
 - What are we looking at?

FinFET Transistors (May 2011 and After)



- FinFETs
 - Why is it called that?
 - What do you think the advantage is?

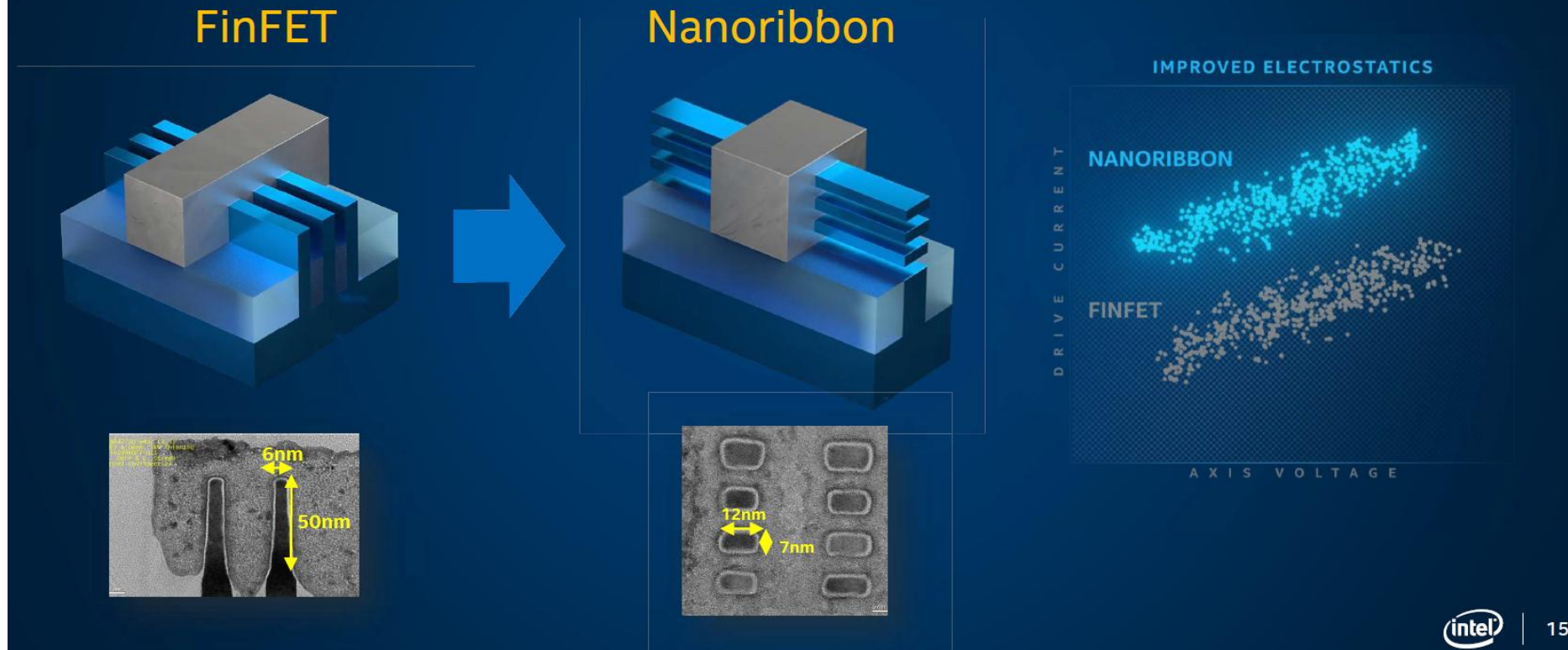
Scaling – How much have we scaled devices?



- This is a recent roadmap for Intel's transistor scaling plans

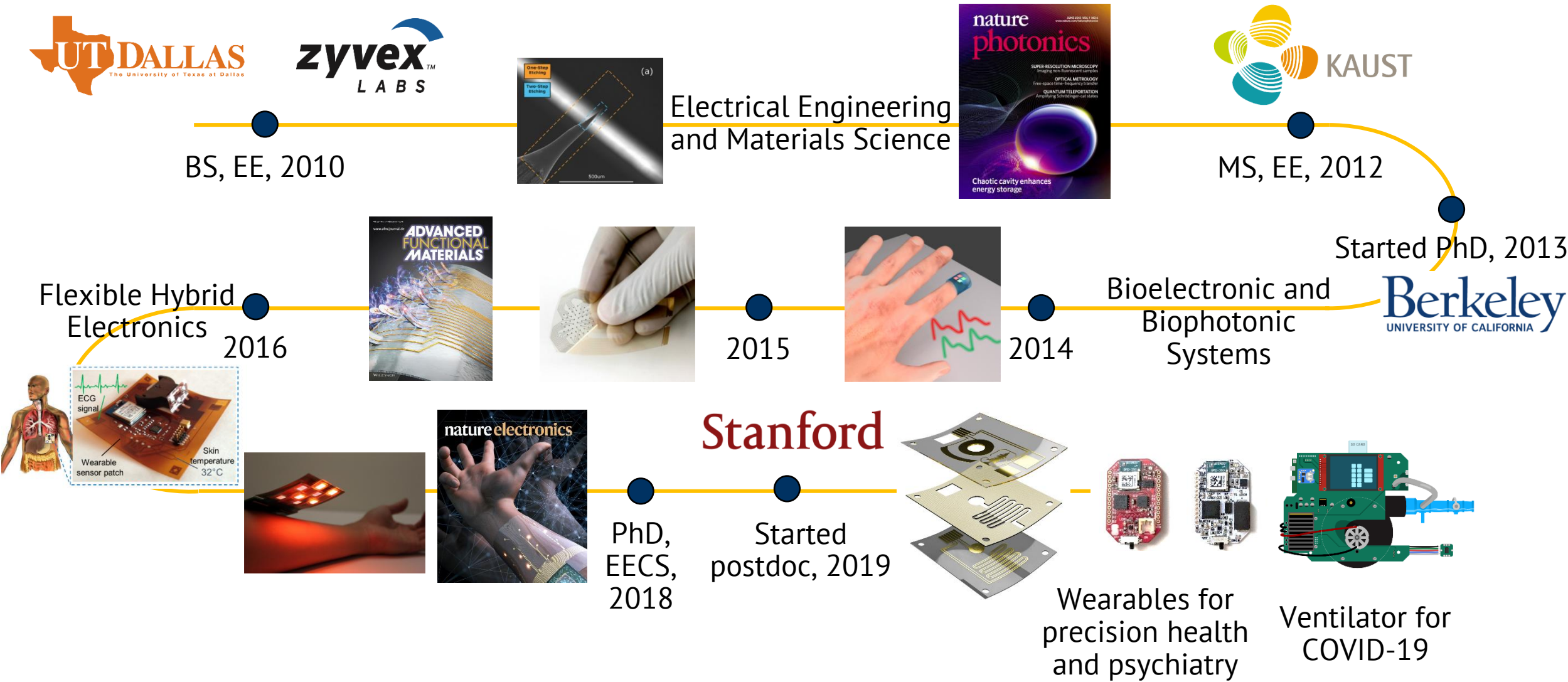
Scaling – How much have we scaled devices?

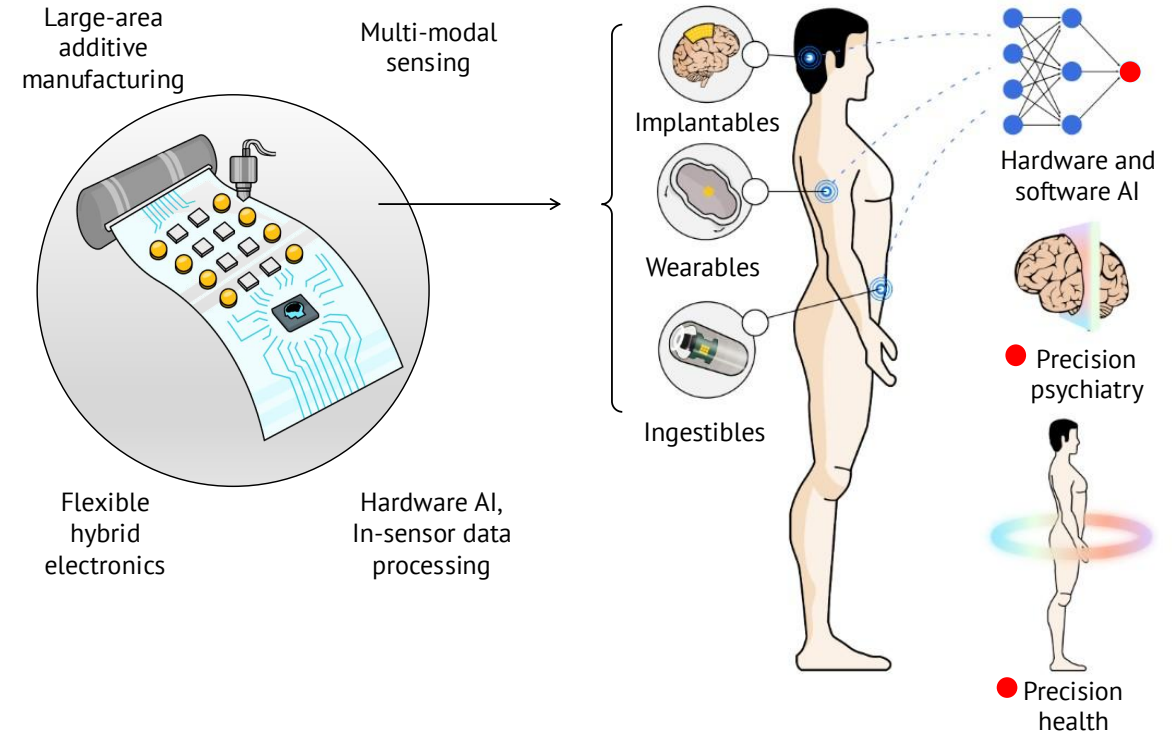
FINFET CHANGING TO NANOWIRE/NANORIBBON

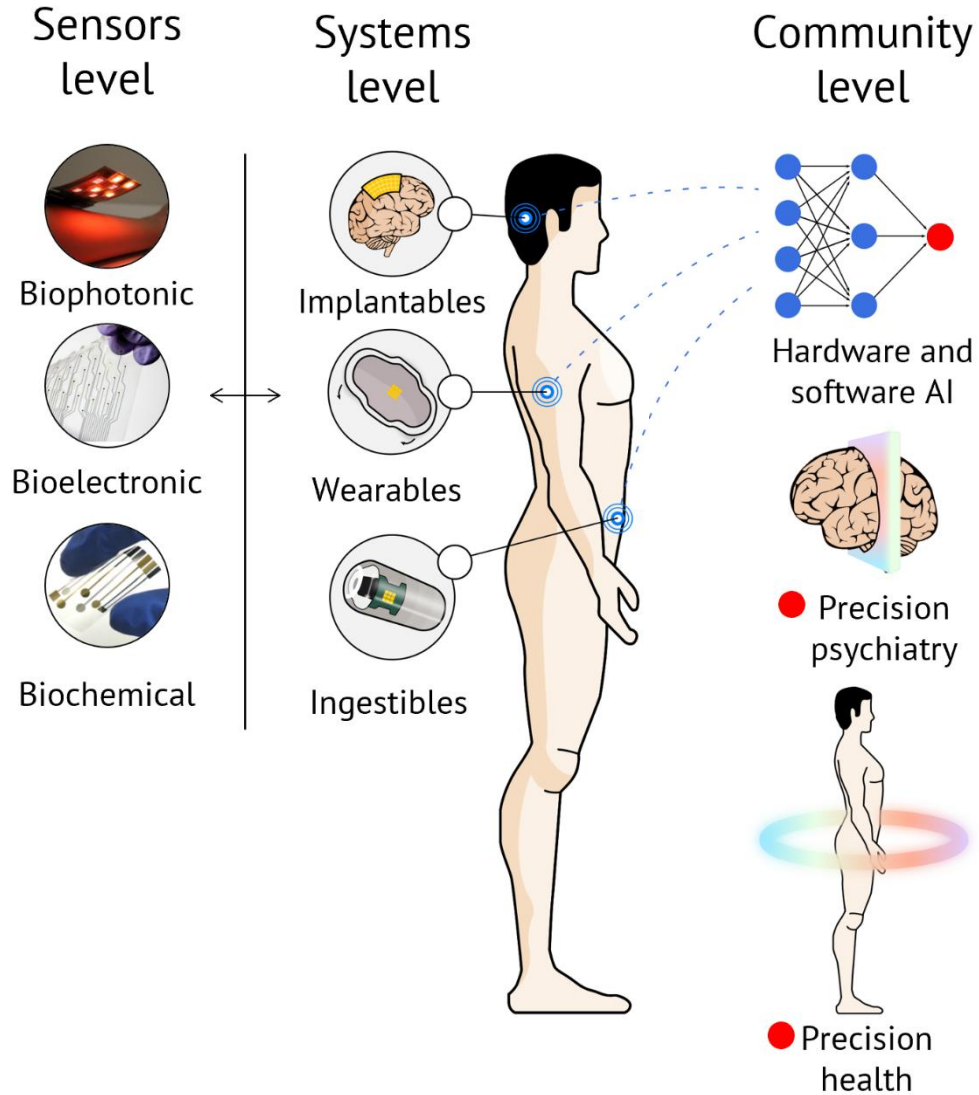


- What is a Nanoribbon FET?
 - Why does it give better performance?

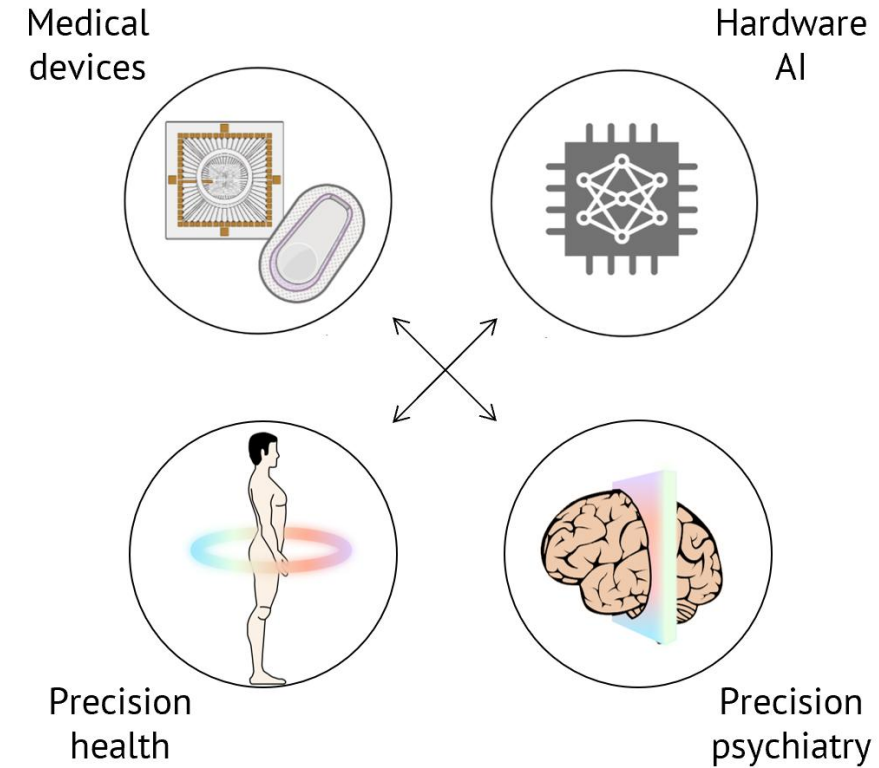
My background and research



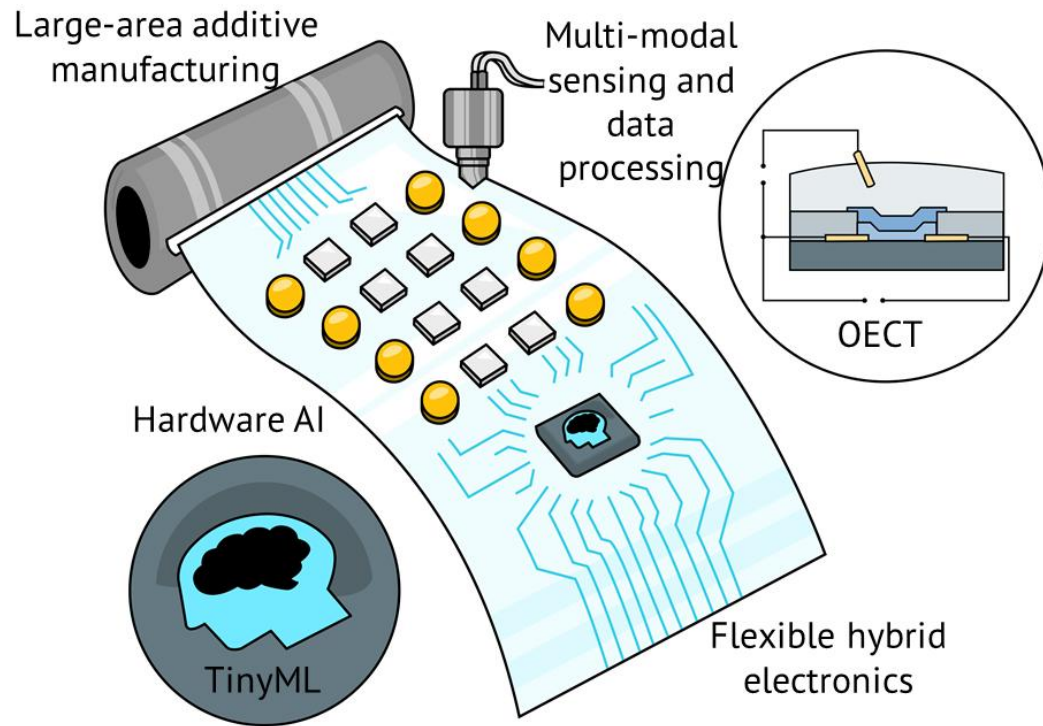




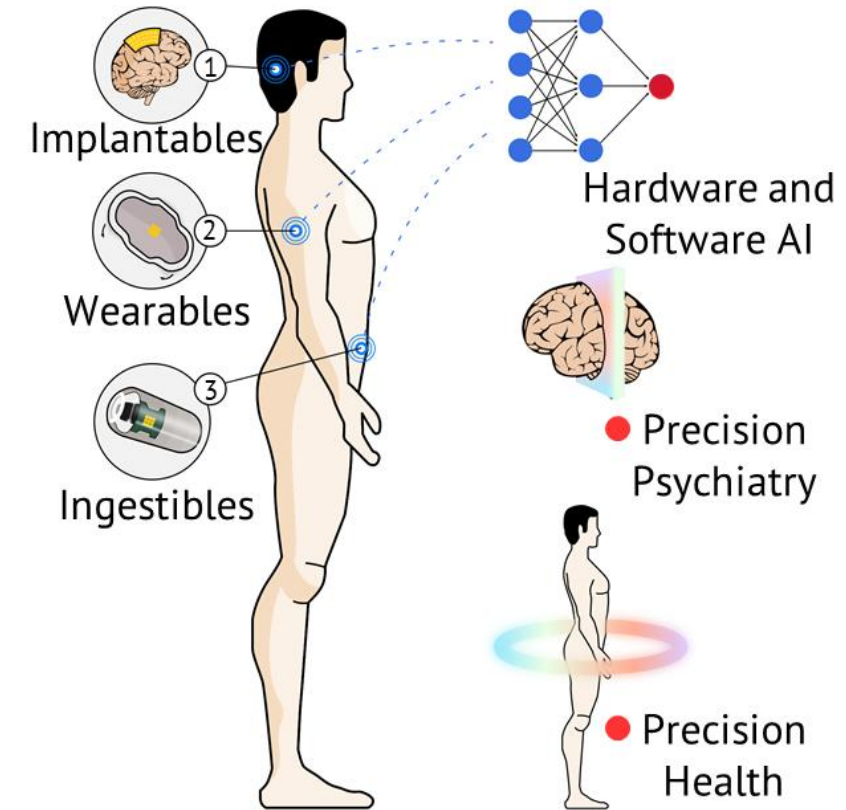
Research Focus



Soft sensor systems for precision health and psychiatry



Materials,
Manufacturing, Design,
Devices, and Circuits



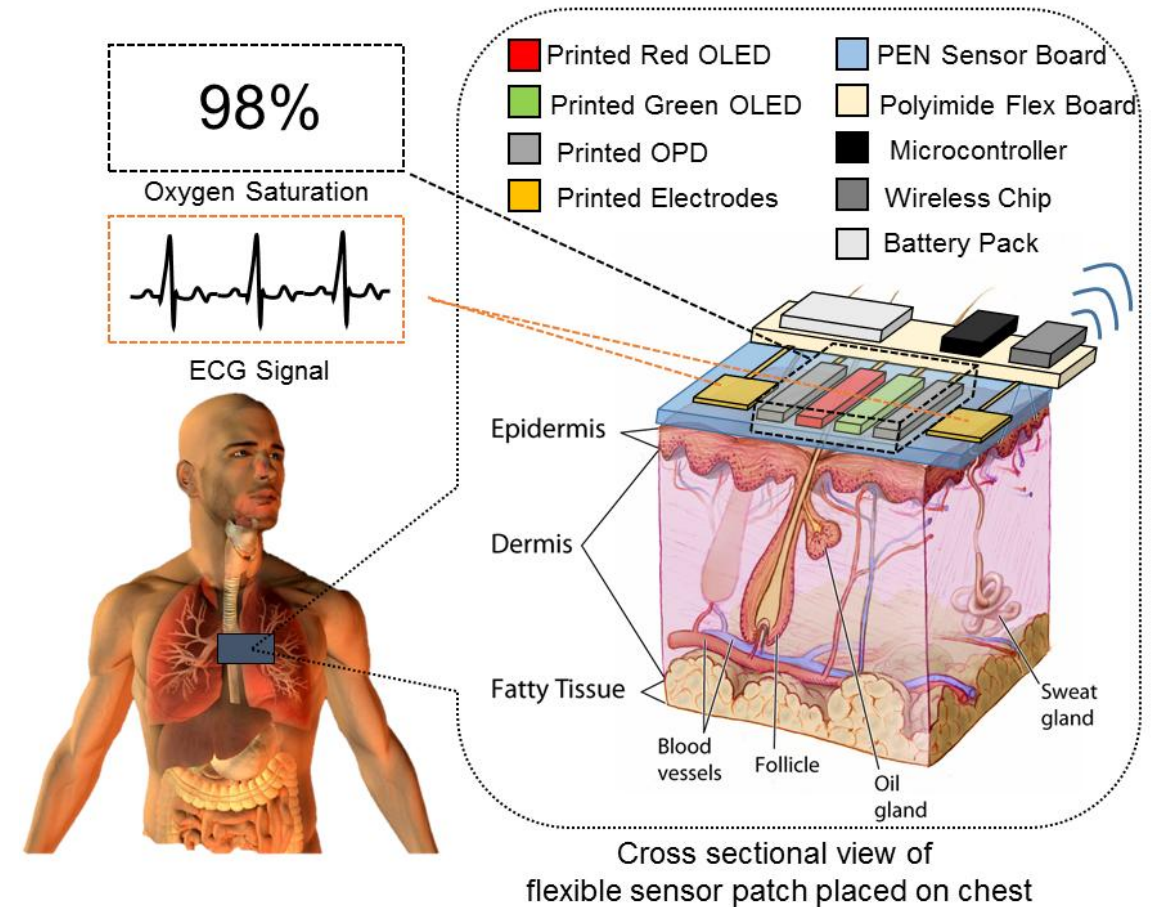
Systems and
Applications

Sensors for precision health and psychiatry



Current state:

Big machines with a lot of wired rigid sensors that limits the comfort of the patient



Vision:

A flexible sensor patch that seamlessly measure vital signs - measurements of the body's most basic functions

AI-powered biosensor for detecting hand gestures

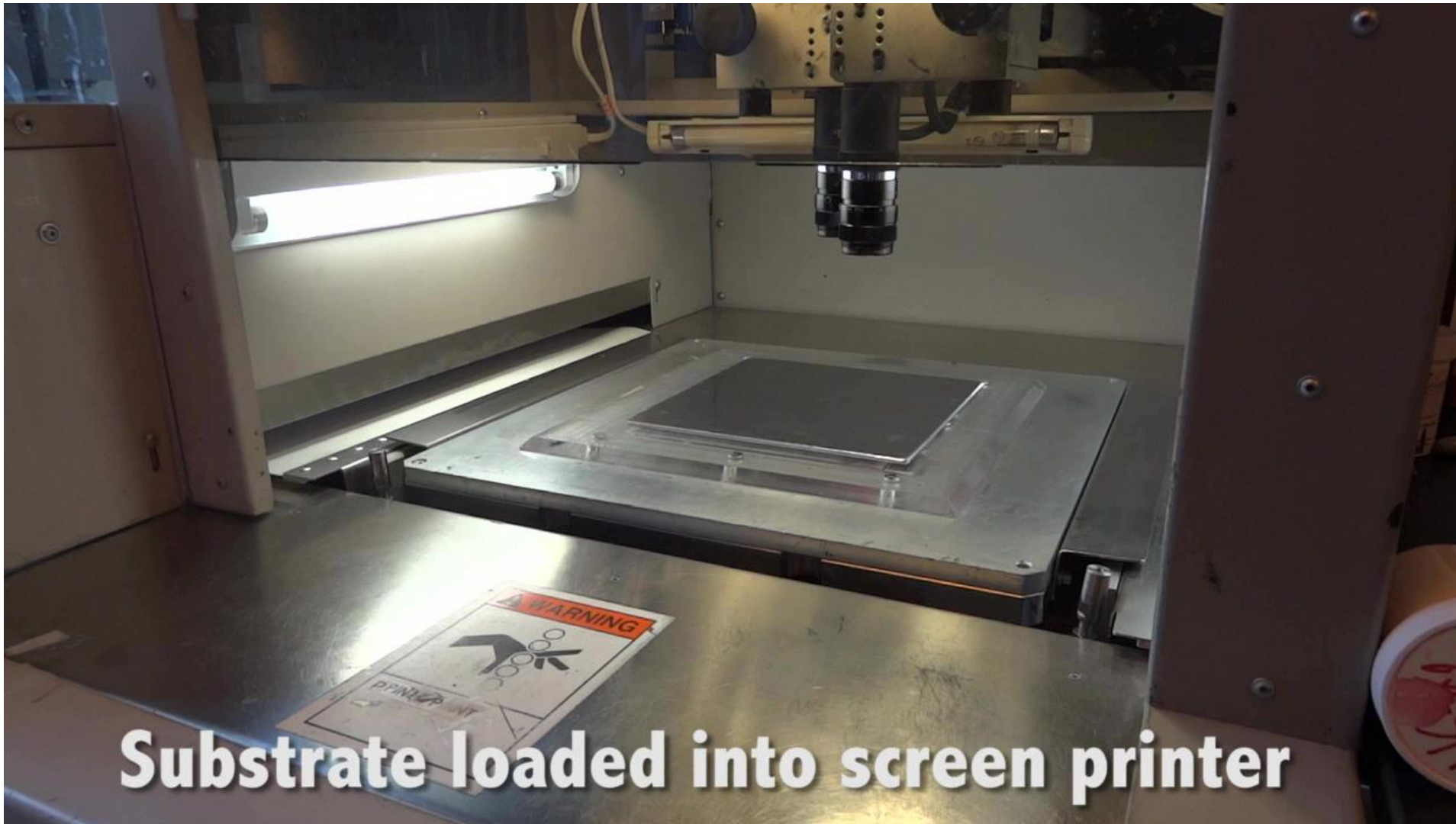
- Biopotential sensing for gesture recognition

FlexEMG

HIGH-DENSITY FLEXIBLE EMG-BASED GESTURE RECOGNITION WEARABLE DEVICE USING
HYPER-DIMENSIONAL COMPUTING

Berkeley
UNIVERSITY OF CALIFORNIA

Screen-printed MRI receive coils

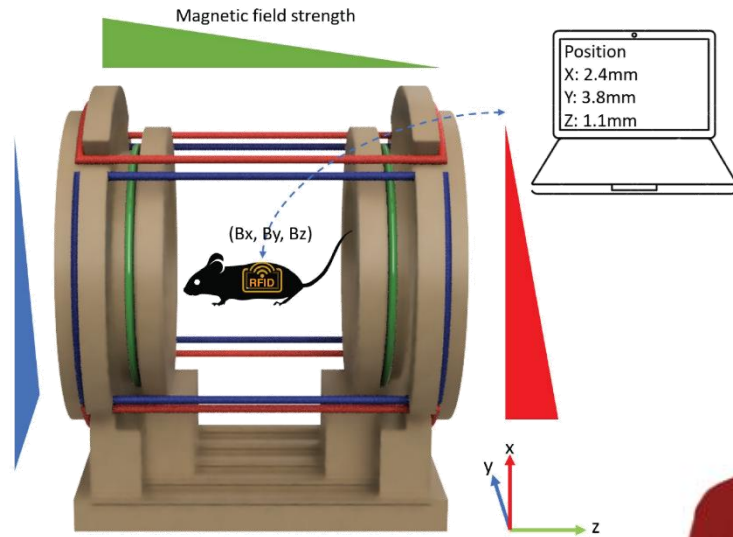


- High-throughput.
- Scalable.
- Fast.
- Significantly less process steps compared to photolithography.

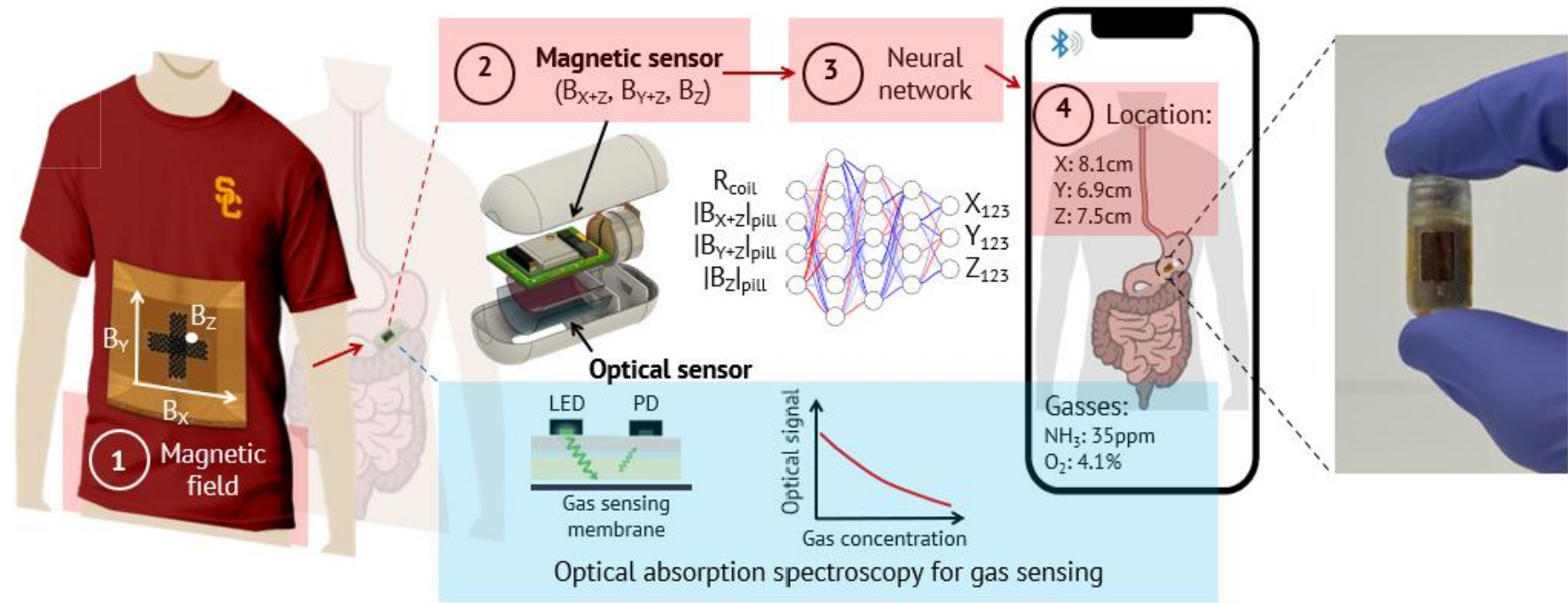
Substrate loaded into screen printer

3D gas mapping in the gut with AI-enabled ingestible and wearable electronics

Existing approach: Helmholtz coil for producing a region of nearly uniform magnetic field and localizing ingestible inside the coil.



Our approach:
Wearable multi-layer
coil for 3D field
gradient, and use a
bend sensor for
accounting for
movement.



3D gas mapping in the gut with AI-enabled ingestible and wearable electronics

From Wearables to Swallowables: USC Engineering Researchers Create GPS-like Smart Pills with AI

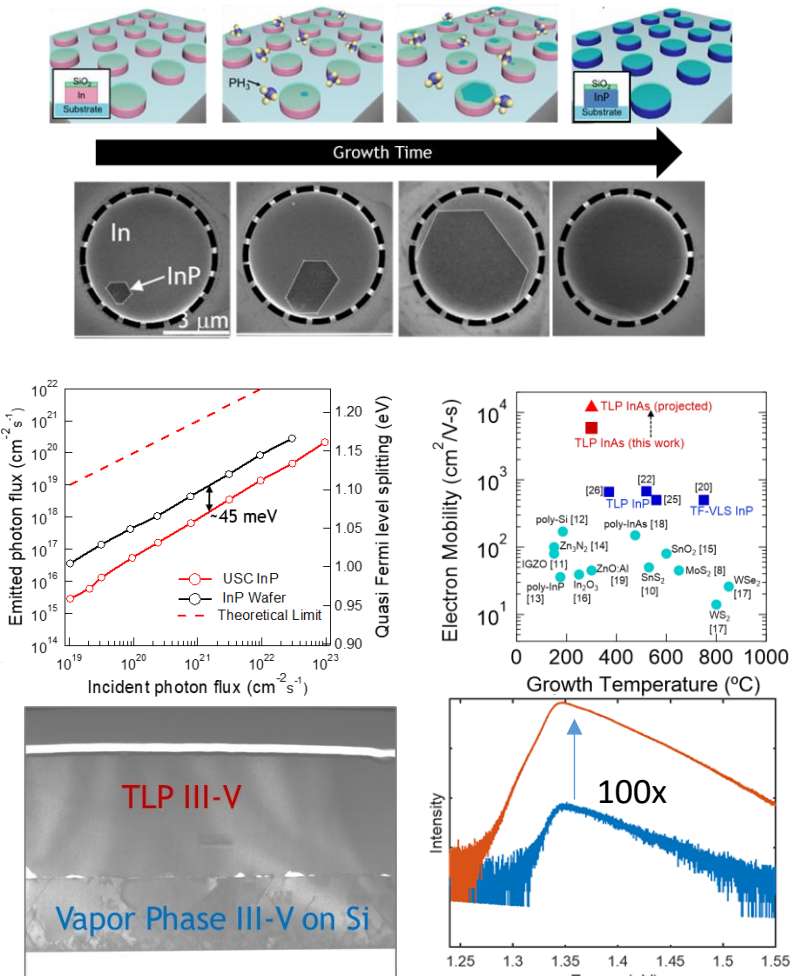
[Amy Blumenthal](#) | June 12, 2024

Utilizing wearable electronics and AI, new ingestible sensors provide real-time 3D monitoring of gastrointestinal health

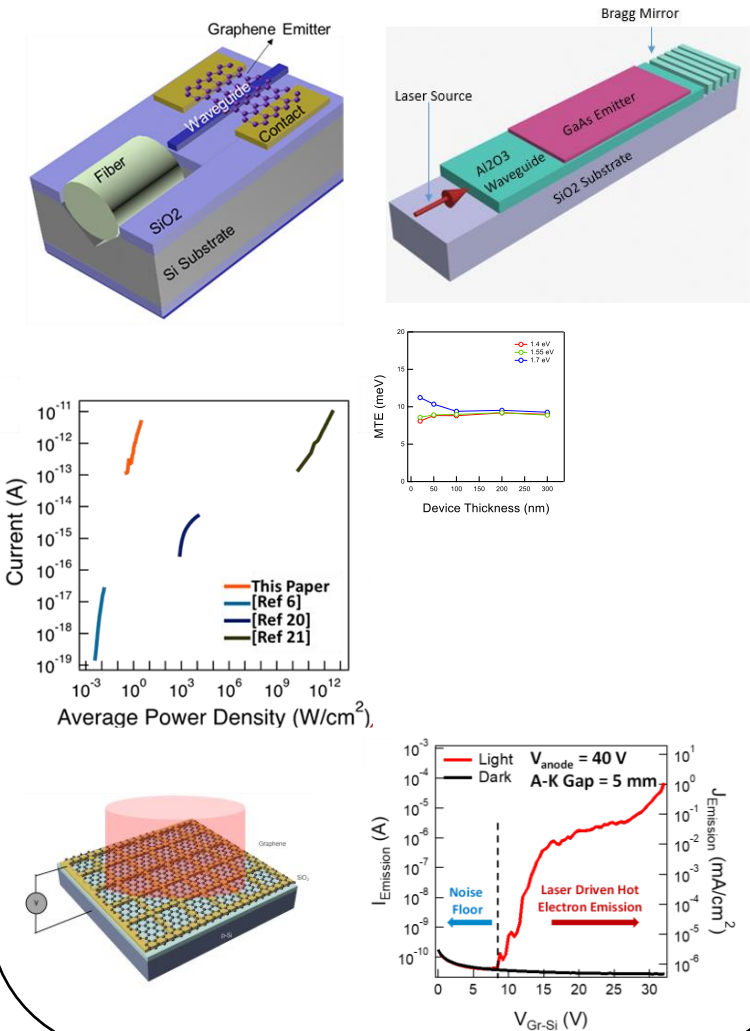


AN INGESTIBLE SMART SENSOR COULD DETECT CANCERS. IMAGE CREDIT: THE KHAN LAB

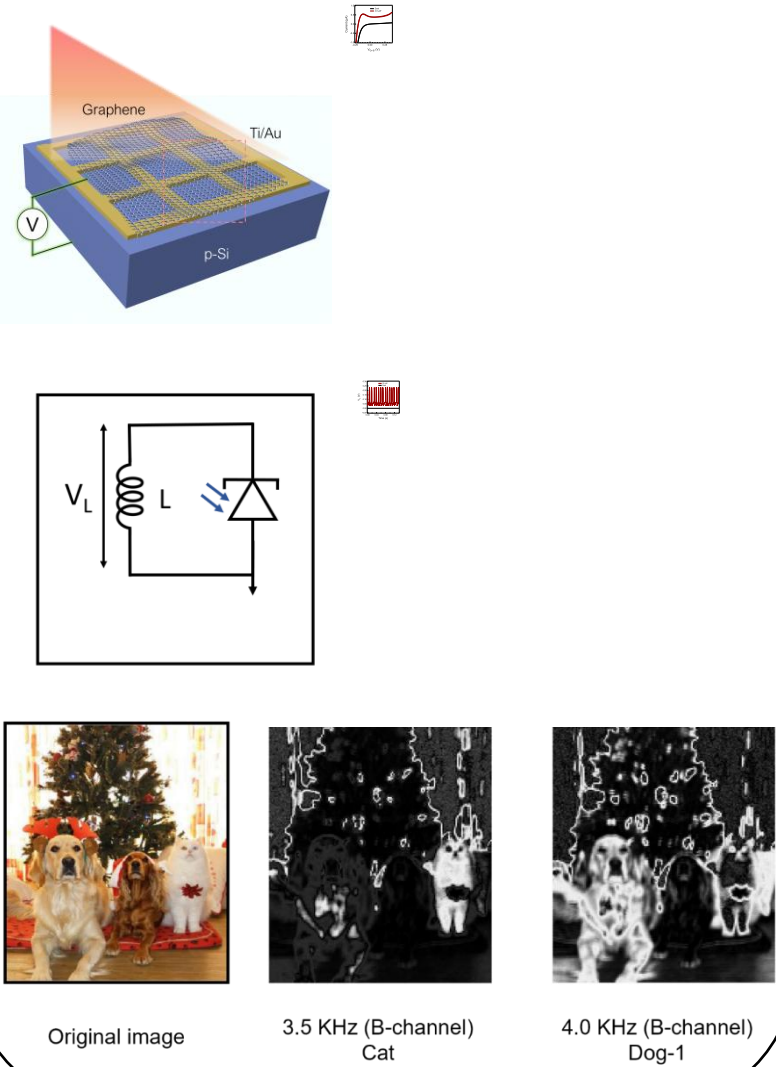
Back-End Materials Growth and Heterogeneous Integration



Electron Emission Devices for GHz to THz Applications



Ultra-Low Power Computing



John O'Brien Nanofabrication Laboratory



Etch

Deposition

Lithography

Wet Processing

Packaging

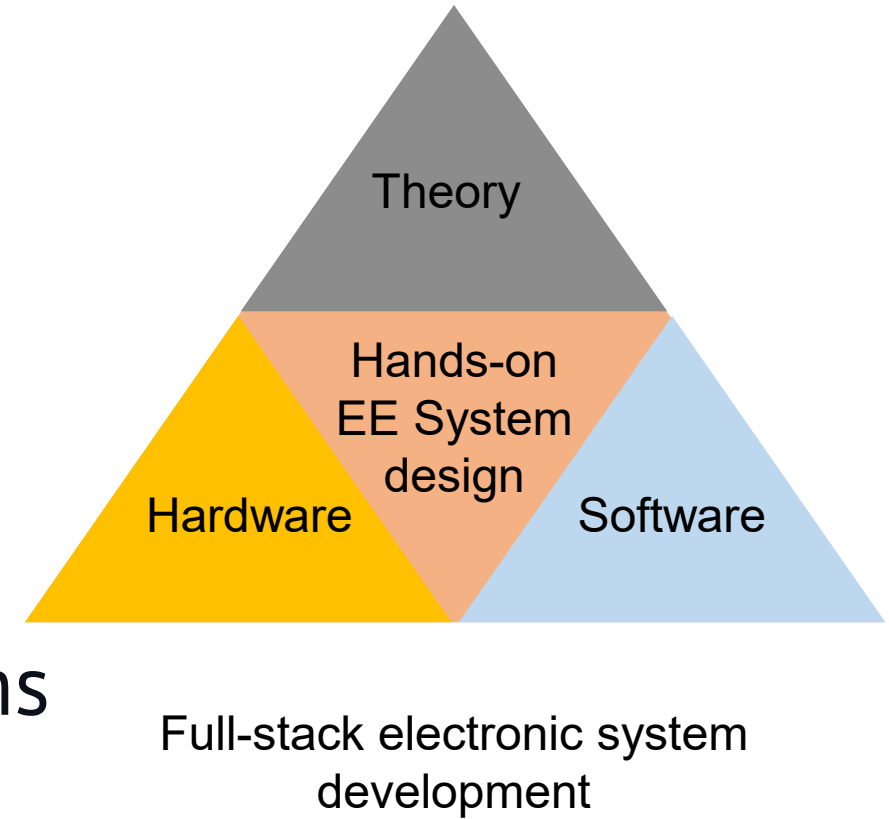
Infrastructure

- \$50M 12,000 Sq. ft. facility
- +24,000 sq. ft. automated facility for facilities, gases, and support systems
- 35+ state-of-the-art cleanroom tools including e-beam and direct write tools, Si and compound semiconductor processing
- Sample size from pieces to 100mm (min) and 150mm (subset – but all III-V)

Scalability:

- +200 users (2024) with expansion to +500 users planned. The lab can handle 100 active users.
- Space available within current facility for additional capacity and capability

- Understand theoretical and practical concepts in electrical engineering.
- Overview of devices, circuits, systems, and AI/ML tools.
- Design simple biosensor systems from end to end.
- Design imager systems with embedded ML.
- Design and implement electronic systems for given specifications.



- Course content and schedule: ee105.org
- Announcement and grading: Brightspace

Instructor



Yasser Khan

yasser.khan@usc.edu

I am a maker, love to build things!

Teaching Assistants



Ibrahim Ozaslan

ozaslan@usc.edu

My research is on system-theoretic framework for designing and analyzing optimization algorithms using tools from robust control theory.



Sushmit Hossain

hossains@usc.edu

Office Hours Somewhere on the internet

My research is on memristors for robotic applications, nanophotonics.



Walter Unglaub

unglaub@usc.edu

I also enjoy teaching/tutoring, app development, entrepreneurship, and learning piano in my free time.

Course structure

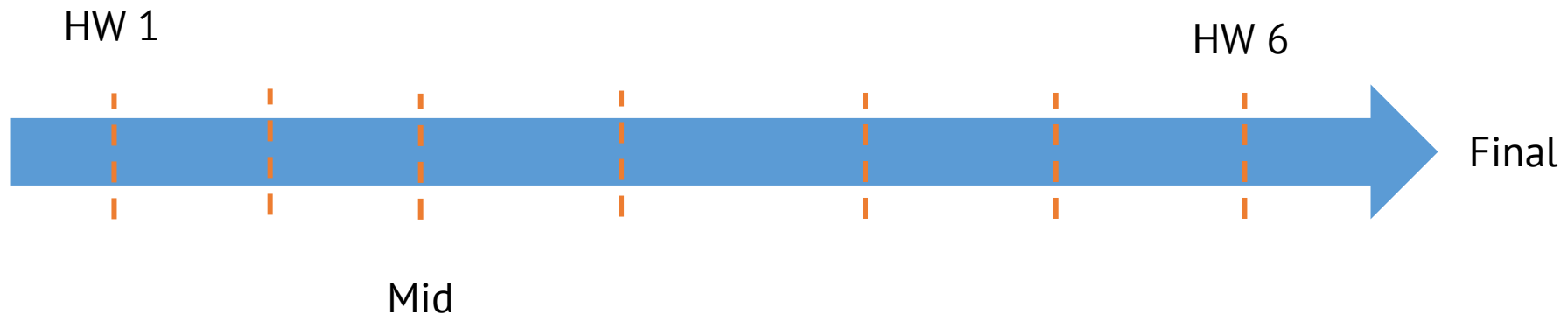
- Lectures and discussions are sufficient for homework and final exam
- Emphasis on *practical learning*
- Homework to review major concepts

Grading Breakdown

Component	Weight	Details
Final Exam	40%	Comprehensive; two pages of handwritten notes allowed (no printing).
Midterm Exam	30%	Covers circuits and devices; one page of handwritten notes allowed (no printing).
Homework	20%	Six assignments: 1. Arduino/Python 2. Circuits 3. Devices 4. Sensors 5. Neural Networks 6. Optical Communications
Final Project	10%	Group project (teams of 5) using demo board and Arduino. • Project ideas due by Oct 23 via email. • Deliverables: 1. Class presentation/2-min youtube video (5 min) – 5% 2. Report (2 pages) – 2.5% 3. Source code (zip file) – 2.5%

Approximate schedule

- Course homepage: ee105.org
- 6 homework assignments
- Final in finals week
- Mid October 7 in class



Course content

- 6 homework are on 6 modules
 - Arduino/python
 - Circuits
 - Devices
 - Sensors
 - Neural network
 - Communication systems
- Midterm will cover circuits and devices.
- Final will cover everything.

Calendar

Module 0 - Introduction to EE 105

Aug 26:	Introduction to Electrical Engineering	Handout the boards
Aug 28:	Introduction to Electronic Systems	LAB Show demo of the board
Sep 02:	LAB Arduino Review	
Sep 04:	LAB iPython Review	
HW 1 RELEASED		

Module 1 - Circuits

Sep 9:	Circuits 1	LAB Symmetric touchpanel
Sep 11:	Circuits 2	LAB Asymmetric touchpanel
HW 1 DUE		
Sep 16:	Circuits 3	TUTORIAL Circuit problems
Sep 18:	Circuits 4	LAB Thermistor lab
HW 2 RELEASED		

Module 2 - Device

Sep 23:	Device 1	TUTORIAL Device problems 1
Sep 25:	Device 2	LAB Diode analysis
HW 2 DUE		
HW 3 RELEASED		
Sep 30:	Device 3	LAB Transistor analysis
Oct 2:	Device 4	TUTORIAL Device problems 2
HW 3 DUE		
Oct 7:	MIDTERM	
Oct 9:	NO CLASS FALL RECESS	

Course content

Module 3 - Sensors

Oct 14:	Sensors 1	TUTORIAL Sensor problems
Oct 16:	Sensors 2	TUTORIAL Op Amp problems
Oct 21:	Sensors 3	LAB PPG
Oct 23:	Sensors 4	LAB Oximeter
FINAL PROJECT PLAN DUE		
HW 4 RELEASED		

Module 4 - Machine Learning

Oct 28:	Nueral network 1	LAB MNIST classification with python
Oct 30:	Nueral network 2	LAB Iimage/text classification with arduino camera
Nov 4:	Nueral network 3	TUTORIAL NN problems
Nov 6:	Nueral network 4	LAB Iimage/text classification with arduino and TinyML
HW 4 DUE		
HW 5 RELEASED		

Module 5 - Communications

Nov 11:	NO CLASS - VETERANS DAY	
Nov 13:	Communications 1	LAB LED/PD combo
Nov 18:	Communications 2	DEMO Quantum comm
Nov 20:	Communications 3	TUTORIAL Noise problems
Nov 25:	Communications 4	LAB Optical comm
HW 6 RELEASED		
HW 5 DUE		
Nov 27:	NO CLASS - THANKSGIVING	

Module 6 - Final Project

Dec 2:	Final project presenations	—
Dec 4:	Final project presenations	—
HW 6 DUE		

Custom demonstration board for EE 105

Circuits

Voltage dividers

Arduino Nano 33 BLE

Arduino

Module 3: Devices

Electronic devices

Diodes

Transistors

Module 2: Circuits

Module 4: Sensors

Electronic systems

Oximeter

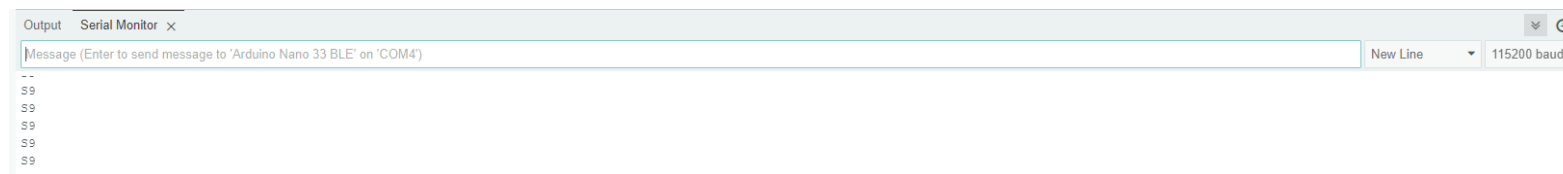
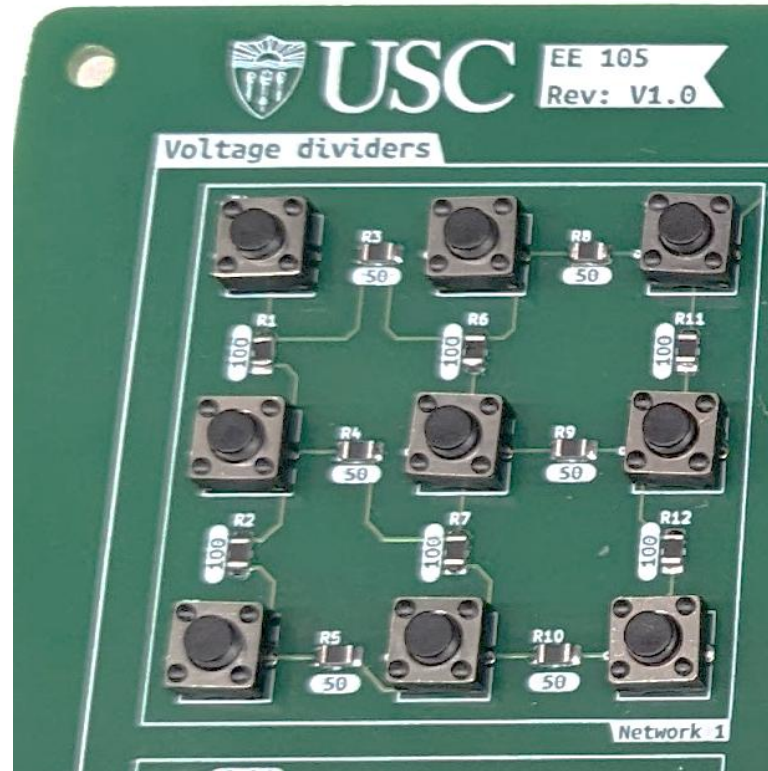
Module 6: Comm

Module 5: NN

Machine vision

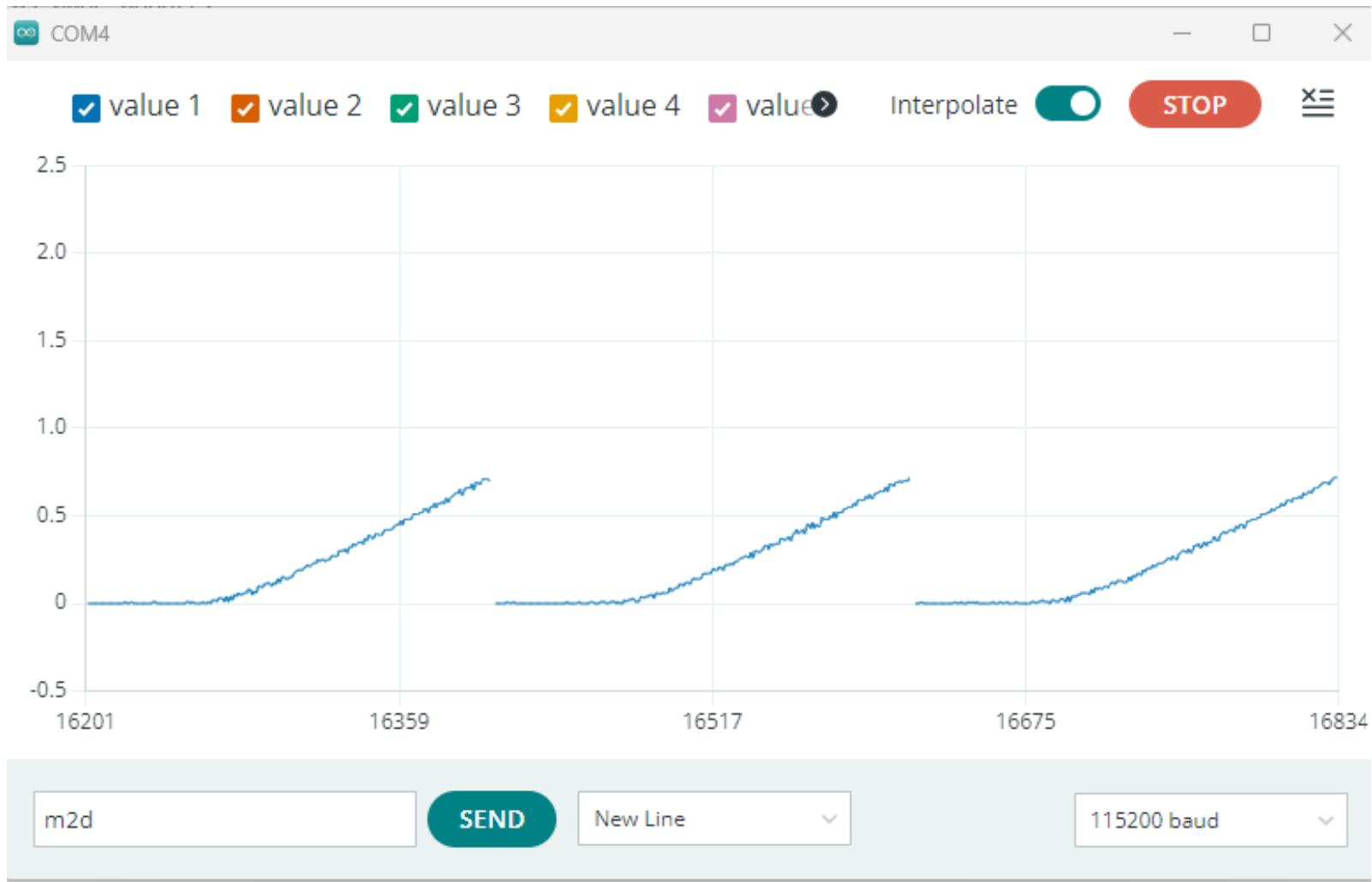
The image shows a custom green PCB demonstration board for EE 105. The board is populated with an Arduino Nano 33 BLE microcontroller. Various electronic modules are integrated, including Module 1 (Network 1), Module 2 (Circuits), Module 3 (Devices), Module 4 (Sensors), Module 5 (NN), and Module 6 (Comm). A camera module is also present. The board includes sections for voltage dividers, power management (M1 Power, M1 Status, M2 Power, M2 Status, M3 Power, M3 Status), and sensor interfacing (Sense Amp, TIA). Callouts identify specific components and modules.

Module 1 on circuits – demo resistive touchscreen



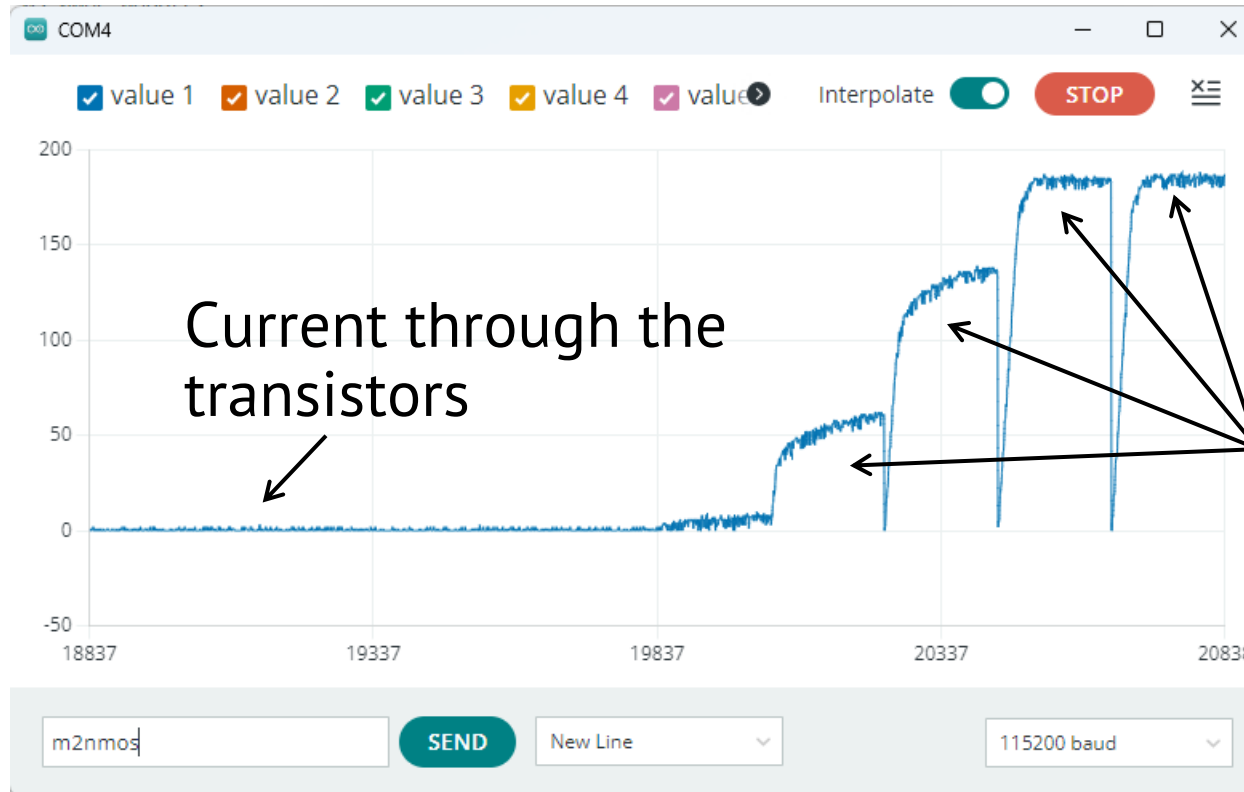
Finds the pressed button location automatically

Module 2: diodes and transistors



Characterize
semiconductor
diode – how do
they work

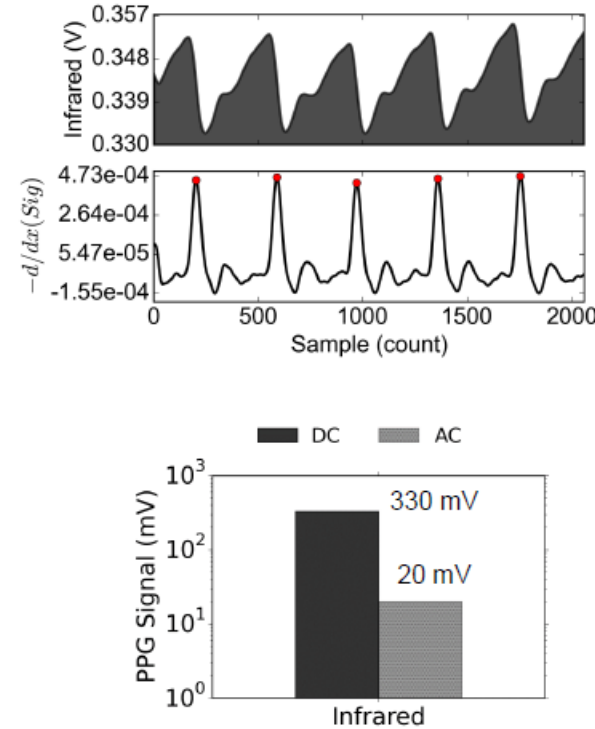
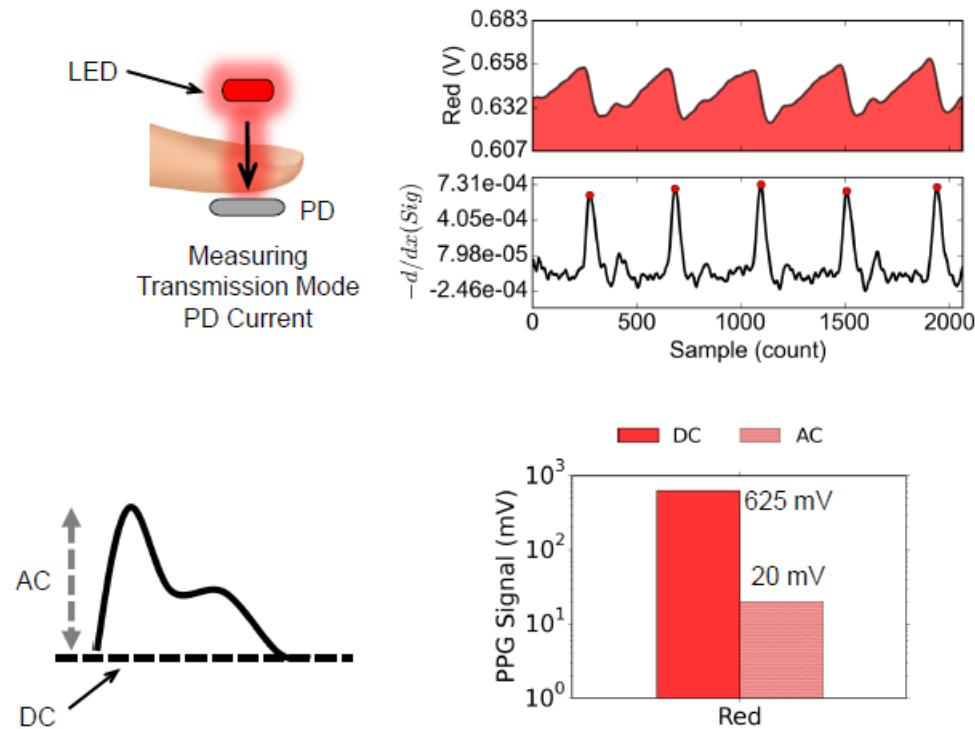
Module 2: diodes and transistors



Characterize
semiconductor
transistors – how
do they work

Increasing gate
voltage / opening
faucet in the water
analogy

Module 3: sensors

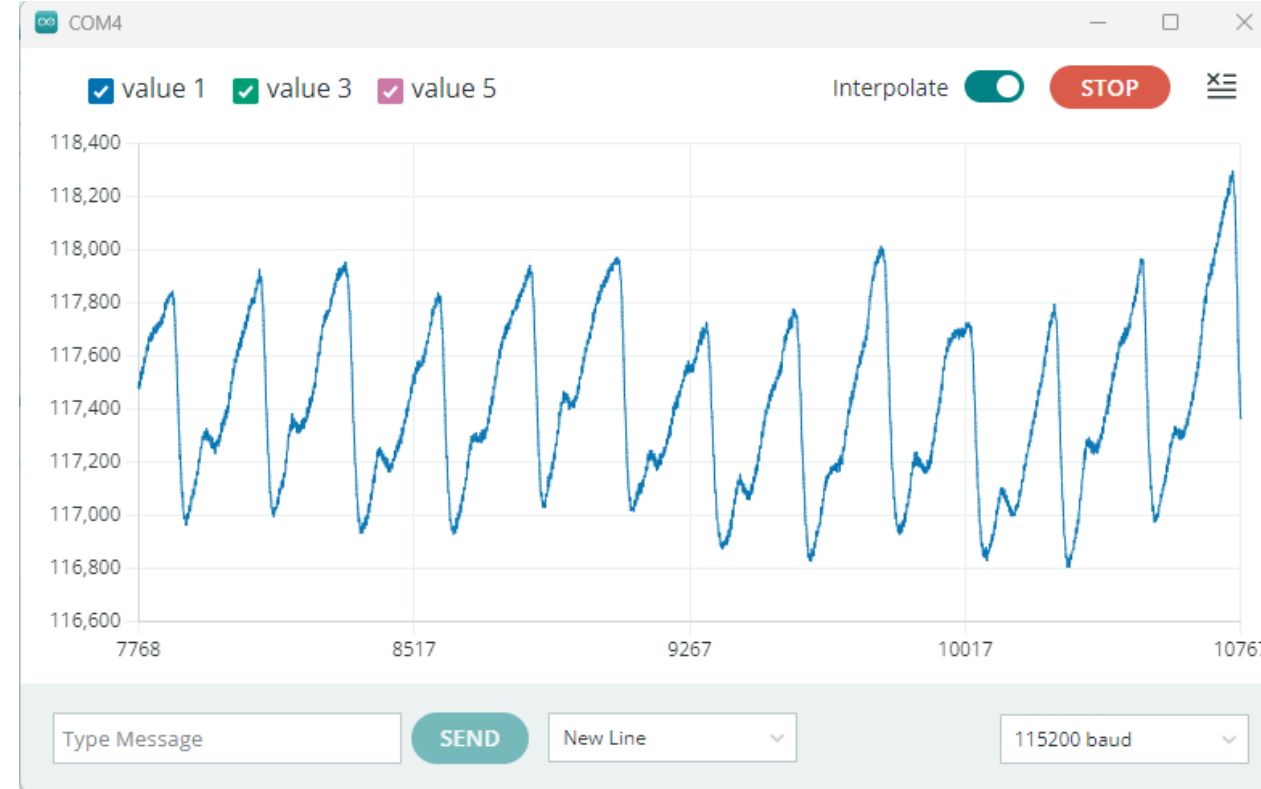


$$R = \frac{AC_{rd}/DC_{rd}}{AC_{ir}/DC_{ir}}$$

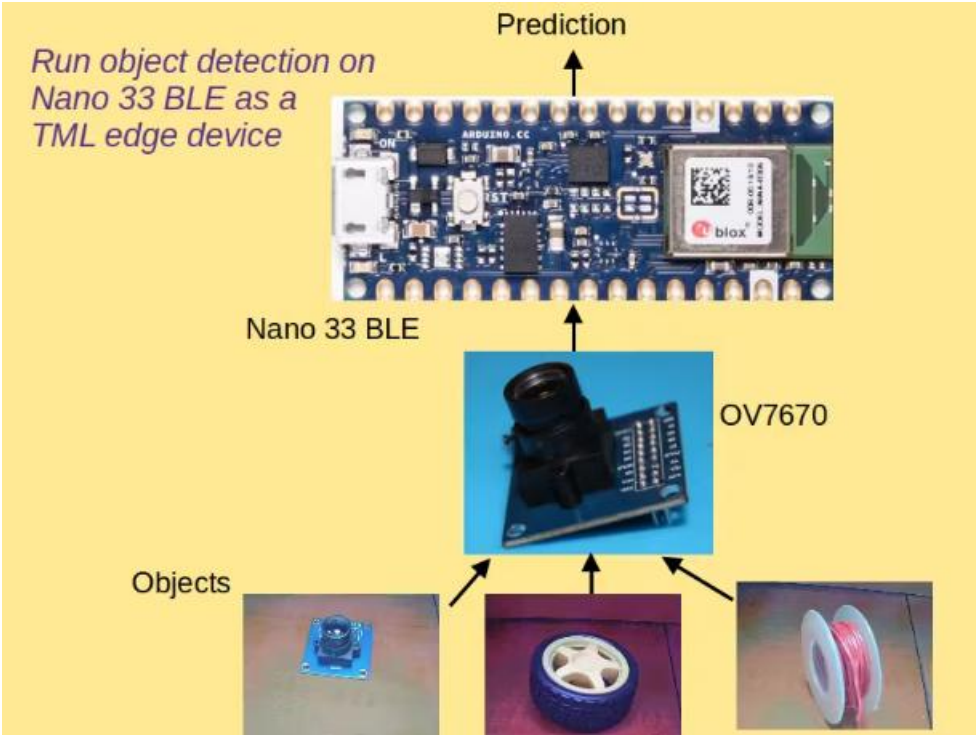
$$R = \frac{20\text{mV}/625\text{mV}}{20\text{mV}/330\text{mV}} = .528$$

$$SpO_2 = 110 - 25 * R = 97\%$$

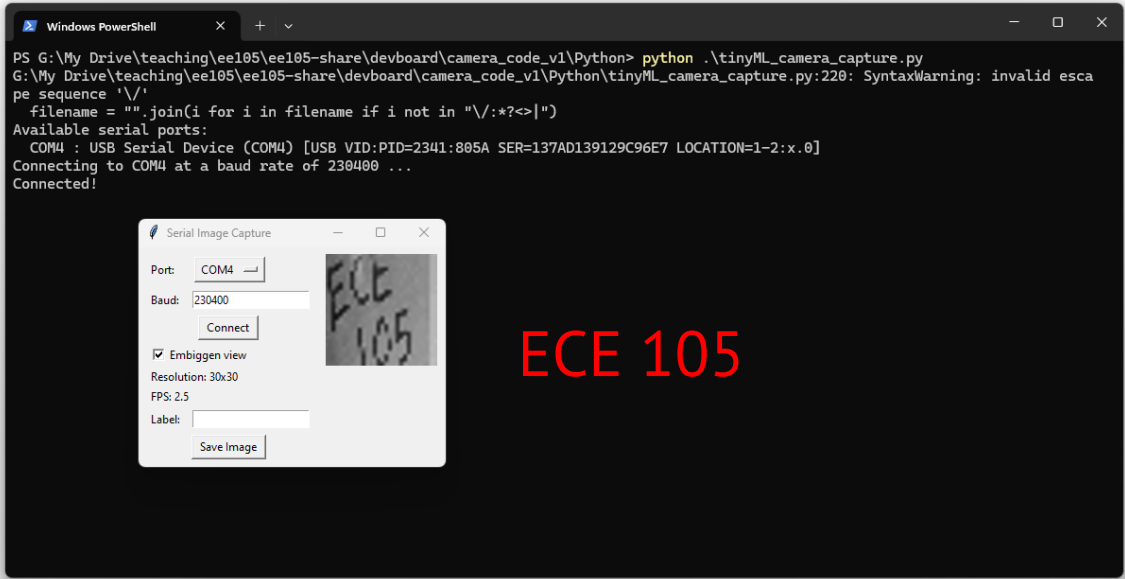
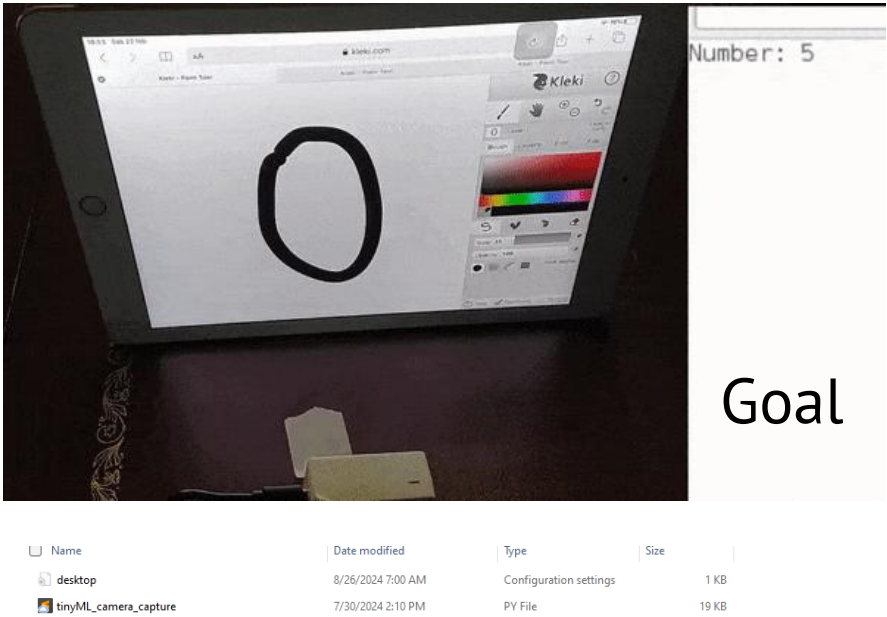
Module 3: sensors - oximeter



Module 4: Machine vision

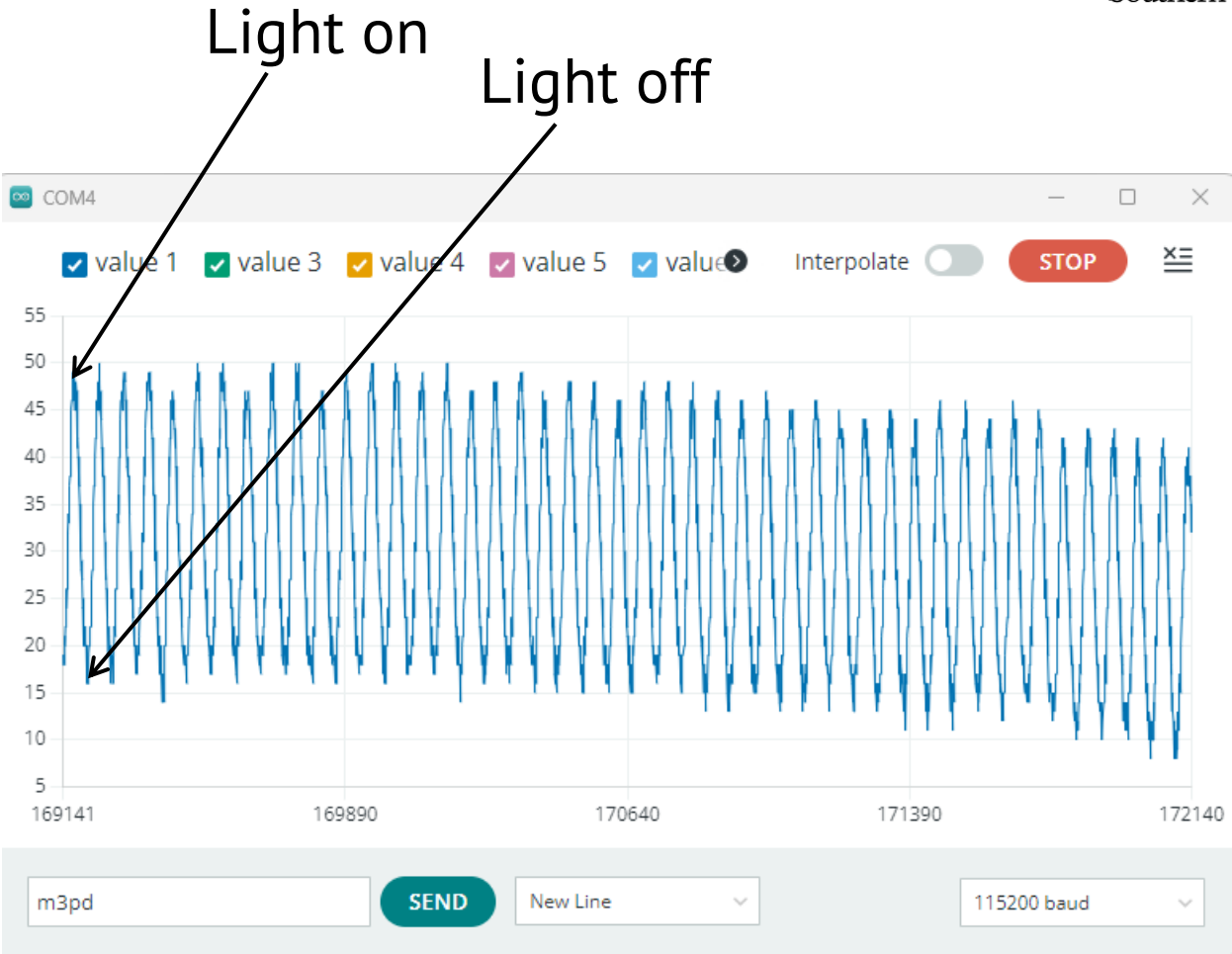
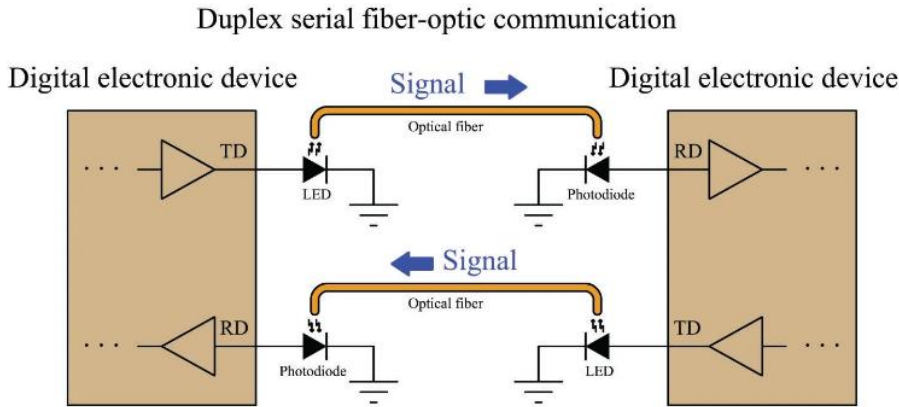


Arduino + camera + ML =
identify objects / letters /
faces / anything



ECE 105

Module 5: optical communication



Next lecture: system design

